

GSFC NEON Series-1 CMO

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Near Earth Orbit Network Series-1 Project, Code 473

**Near Earth Orbit Network
Project Series-1
Sounder for Microwave-Based
Applications
Contract Data Requirements List**



**Goddard Space Flight Center
Greenbelt, Maryland**

**Near Earth Orbit Network Project Series-1
Sounder for Microwave-Based Applications
Contract Data Requirements List
Review/Signature/Approval Page**

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Preface

This document is under NEON Series-1 (NS1) Project configuration control. Once this document is approved, NS1 approved changes are handled in accordance with Class I and Class II change control requirements as described in the Low Earth Orbit (LEO) Programs Division (LPD) Programs and Projects Configuration Management Procedure (CMP), and changes to this document shall be made by complete revision.

Any questions should be addressed to:

LEO/NEON Configuration Management Office
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Greenbelt, MD 20771

Change History Log

Revision	Effective Date	Description of Changes
Revision –	December 20, 2024	This version incorporates 470-CCR-24-0404 which was approved by the SMBA ERB on March 26, 2024, and incorporates NS1-CCR-0004 by the NS1 SMBA CCB on the effective date shown.
Revision A	April 1, 2025	This version incorporates NS1-CCR-0007 which was approved by the NS1 SMBA CCB on the effective date shown. Removed duplicate language for UAR. Changed delivery date for MA-1.
Revision B	August 19, 2025	This version incorporates NS1-CCR-0014 which was approved by the NS1 SMBA CCB on the effective date shown.

Table of Contents

1	INTRODUCTION	1
1.1	Scope.....	1
1.2	Export Control Limitations	1
1.2.1	ITAR Technical Data	1
1.2.2	EAR Technical Data.....	1
1.3	Proprietary Data	2
1.4	Prior Submission Deliveries	2
1.5	Due Dates – Related Definitions.....	2
1.6	Media – Related Definitions	3
1.7	Category – Related Definitions.....	4
1.8	Applicability	4
1.9	Updates	5
1.10	Documentation Change Procedures.....	5
1.11	Delivery Instructions.....	5
2	DATA ITEM DESCRIPTIONS	19
2.1	Calibration/Validation Data Items	19
2.2	Integration and Test Data Items.....	32
2.3	Mission Assurance Data Items.....	40
2.4	On Orbit Data Items.....	51
2.5	Project Management Data Items.....	64
2.6	Review Data Items.....	77
2.7	Software Data Items.....	103
2.8	Systems Engineering Data Items	113
	APPENDIX A: ABBREVIATIONS AND ACRONYMS	147

List of Tables

Table 1.	Contract Deliverable Items List.....	7
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1 INTRODUCTION

This contract data requirements list (CDRL) document defines the requirements for deliverable documentation to be provided by the Sounder for Microwave-Based Applications (SMBA) Contractor. Section 1 includes definitions and instructions for distribution. Table 1 presents the CDRL item by item, with due dates, quantity, and media format. Section 2 contains the data item descriptions (DID), descriptions of each item which provide the use for the deliverables and the required preparation information. Except where specifically indicated to the contrary, the formats and drawing standards used shall be those normally used by the Contractor and by its subcontractors.

1.1 Scope

The CDRLs within this document include those required for Phase I, the development phase, as well as Phase II, the production phase, as indicated in Table 1. Though this document has been created for the Near Earth Orbit Network (NEON) Series-1 project, the CDRLs, particularly those associated with Phase II, are applicable to subsequent projects withing the NEON Program.

1.2 Export Control Limitations

Some of the content for documents provided by the National Aeronautics and Space Administration (NASA) (Goddard Space Flight Center [GSFC]) and United States (U.S.) industry will be subject to the U.S. International Traffic In Arms Regulations (ITAR), 22 CFR 120-130, by the U.S. Department of State and/or the Export Administration Regulations (EAR), 15 CFR 730-744, by the U.S. Department of Commerce. NASA may exchange technical information with the foreign partners and related entities (e.g., contractors and subcontractors) utilizing licensing exemptions to the ITAR and/or EAR, supported by the authority and content of the standing Letter of Agreement (LOA) and/or the Memorandum of Understanding (MOU) between NASA/the National Oceanic and Atmospheric Administration (NOAA) and that foreign partner, as appropriate. U.S. industry may exchange ITAR sensitive technical data under the authority and conditions of Technical Assistance Agreements (TAA), approved by the U.S. Department of State as appropriate.

NASA and U.S. industry will mark and thus notify foreign partners and related entities of the exchange of ITAR and/or EAR sensitive data with variations of the following language.

1.2.1 ITAR Technical Data

The unclassified technical information included is controlled under the U.S. ITAR, 22 CFR 120-130, by the U.S. Department of State. Transfer of this information to a foreign person or entity requires an export license issued by the U.S. Department of State or an ITAR exemption to the license requirement prior to the export of transfer.

These items are licensed by the United States for ultimate destination [foreign partners and related entities]. Diversion, retransfer, disclosure, or use contrary to applications specified in NASA/NOAA agreement without prior U.S. authorization are prohibited.

1.2.2 EAR Technical Data

The unclassified technical information included is controlled under the U.S. EAR, 15 CFR 730-744, Export Control Classification Number 9E515.a and 9E515.x by the U.S. Department of

Commerce. Transfer of this information to a foreign person or entity requires an export license issued by the U.S. Department of Commerce or an EAR exemption to the license requirement prior to the export of transfer.

These items are licensed by the United States for ultimate destination [foreign partners and related entities]. Diversion, retransfer, disclosure, or use contrary to applications specified in NASA/NOAA agreement without prior U.S. authorization are prohibited.

1.3 Proprietary Data

Some of the content of documents may be of a proprietary nature to the document preparing/sending Party. In the event data is deemed to be proprietary, and for which protection is to be maintained, the sending Party shall mark the document with a notice to indicate that the data therein is proprietary and shall be used and disclosed by the receiving Party and its related entities (e.g., contractors and subcontractors) only for the purposes of fulfilling the receiving Party's responsibilities under the Near Earth Orbit Network (NEON) program, and that the identified and marked technical data shall not be disclosed or retransferred to any other entity without prior written permission of the document preparer. Data improperly marked as proprietary shall be rejected by the contracting officer's representative (COR) and returned for removal of proprietary markings.

1.4 Prior Submission Deliveries

If new hardware, software, or documentation is developed, procured, designed, fabricated, tested, or processed in a different manner from that of the first SMBA flight instrument effort (obsolescence, corrective actions, modifications, errors, etc.), the Contractor shall deliver the data item as required with the necessary updates for Government acceptance.

1.5 Due Dates – Related Definitions

The following definitions apply to the "DUE DATE, MATURITY" column in Table 1. Unless otherwise specified, deadlines are in calendar days.

(a) DUE DATE:

- (1) IOC: Initial Operating Capability
- (2) dCDR, PER, etc.: Delta Critical Design Review (dCDR), Pre-Environmental Review (PER), Pre-Ship Review (PSR), etc. (all major reviews). Electronic distribution to be delivered to the Government five working days prior to review, unless otherwise stated.
- (3) Monthly: Submitted on monthly basis
- (4) T: Launch Date
- (5) DACA/(D): (Calendar) Days After Contract Award/(Definitization)
- (6) JCT#: Joint Compatibility Test (JCT). Several JCTs are planned (reference the statement of work [SOW]), the number indicates for which JCT the deliverable is required (JCT#1 is the first JCT, JCT#2 is the second, etc.).

- (7) MR: Mission Rehearsal (MR). Several MRs are planned (reference the SOW); the number indicates for which JCT the deliverable is required (MR#1 is the first MR, MR#2 is the second, etc.).
- (8) FQT: Formal Qualification Test
- (9) ATP: Authority to Proceed with the contractual effort (definitization or award of letter contract whichever is earlier).
- (10) SCCDR: Spacecraft CDR
- (11) SCPSR: Spacecraft PSR
- (b) MATURITY:
 - (1) Initial: The first submission of an item, which shall be revised and resubmitted at a later date.
 - (2) Preliminary/Prelim: An early submission of an item. To be completed as is practicable at the time of preparation. Preliminary submittals are written with the best available current information and are resubmitted when further information becomes available.
 - (3) Final: The complete, thorough submission of an item for approval, review, or information that, to the best of the Contractor's knowledge and intention, will not require further revision or updates. However, this does not preclude updating if later found to be necessary. Any updates shall require the same "approval/review" process as was required for the original submissions.
 - (4) Current: The delivery is written with the best up-to-date information available at the time.
 - (5) Update: The delivery is revised with the best up-to-date information available at the time.

Other entries in the "DUE DATE" and "MATURITY" columns are self-explanatory.

1.6 Media – Related Definitions

The following definitions apply to the deliverables in Table 1.

E – Data items shall be delivered in electronic format to a GSFC Low Earth Orbit (LEO) Programs Division (LPD) specified web portal unless otherwise noted in the CDRL DID. The Contracting Officer shall be notified of electronic submission of the deliverable via email. Electronic deliverables shall be delivered in the following formats unless otherwise approved by the Government:

Text Documents:	PDF (searchable) or Microsoft Word
Presentations:	PDF (searchable) or PowerPoint
Spreadsheets:	Microsoft Excel
Database:	Delimited ASCII files accompanied with database schema document defining tables and entries.
Schedules:	PDF and Microsoft Project
Schematics and Drawings:	Design Web Format (DWF) and PDF
Photographs:	JPEG or current industry standard.
Video:	Any readily available open standard (e.g., AVI, MPEG)
	Test
Measurement Data	Delimited ASCII format or other acceptable format

In addition to deliveries in the required formats above, the contractor may also provide additional copies of the same deliverables in other formats (e.g., .mdzip).

If separate distribution instructions are not specified for separate submission due date/maturity items, then the listed distribution applies to all submissions.

1.7 Category – Related Definitions

The following definitions apply to the "Submission Category (CAT)" column in Table 1. If separate approval instructions are not specified for separate submission due date/maturity items, then the listed approval instruction applies to all submissions.

- A Approval: CDRL items in this category require approval by the GSFC COR or Contracting Officer prior to use by the Contractor. Receipt by the Government shall occur within the time specified in the "Due Date" column of Table 1 of this document. Requirements for resubmission shall be as specified by the Contracting Officer. For most cases the Contractor shall be required to resubmit the document within 14 calendar days of receiving comments from the Government. If the Contractor has not received response from GSFC within 30 calendar days of delivery of a CDRL item, the Contractor may proceed as if the document has been approved.
- R Review: Documents in this category require delivery to the Government prior to use and within the time period specified in the "Due Date" column of Table 1 of this document. They are subject to evaluation by the Government or its designated representatives to determine Contractor effectiveness in meeting contract objectives. Upon submission, the Contractor may proceed with associated work while the Government reviews the submission. Requirements for resubmission shall be as specified by the Contracting Officer.
- I Information: Data in this category shall be delivered to the Government within the time period specified in the "Due Date" column of Table 1 for the purpose of determining current program status, progress, and future planning requirements.

1.8 Applicability

The following definitions apply to "Applicability (APPLIC)" column in Table 1.

ETU – Applies to the Engineering Test Unit (ETU). This is associated with the Phase I (development) contract.

FM1D – Applies to the first flight model (FM) development. This is associated with the Phase I (development) contract.

FM1P – Applies to the first flight model production. This is associated with the Phase II (production) contract.

AFM – Applies to additional flight models (FM2 on). This is associated with the Phase II (production) contract.

1.9 Updates

If the Government determines that there is a substantive change or update needed to properly reflect the information previously delivered, the contractor shall make the update and redeliver the CDRL independent of the quantity requirements in the CDRL table. The Contractor shall notify the Government of any pending updates and re-deliver the CDRL on a schedule agreed to by the Government. Table 1 does not specify this repetitious requirement, however it is required for each CDRL.

1.10 Documentation Change Procedures

Unless otherwise defined in the SOW or contract:

- a. When a minor correction/change is needed or has been made to a few parts of a “Final” version of a CDRL item that has already been delivered to the Government, the Contractor shall issue within 30 days a Documentation Change Notice starting with the number 001. Change bars or “change tracking” shall be used to indicate the locations of changes unless the particular file type does not have those features available. If not available, the details of the changes will be described in a text file.
- b. When a major correction/change is needed or has been made to all or multiple parts of a “Final” version of a CDRL item that has already been delivered to the Government, the Contractor shall deliver within 30 days a complete revision of the CDRL item. A description of all changes incorporated into the new revision shall be included in the new document or otherwise documented. No change bars are used when a document is updated by revision, and the Documentation Change Notice numbers for that document shall be automatically recycled to 001.

1.11 Delivery Instructions

- a. The Contractor shall deliver each CDRL item via the media indicated in accordance with the schedule and any special instructions indicated in Table 1 and/or the corresponding DID, and in accordance with the contract or SOW requirements and provisions.
- b. All CDRL items shall be delivered electronically (one electronic copy for each CDRL item) to an electronic library designated by the Government. If all or a portion of a CDRL item is not well-suited for electronic delivery, the item may be delivered via hardcopy upon mutual agreement between the Contractor and the Government.

With each prior delivery submission, the Contractor shall include a cover letter with the following information:

- CDRL #
- Title
- Due Date
- Flight Model
- Serial Number
- Original Submission Date
- Prior Delivery CDRL #
- Current Submission Revision
- Current Submission Revision Date

The Contractor may deliver a single letter for multiple submissions as long as the information above is contained in the letter for each deliverable item.

Table 1. Contract Deliverable Items List

No.	CALIBRATION/VALIDATION	DUE DATE	CAT	APPLIC
CV-1	Calibration/Validation and Intercalibration Plan	- Draft: PDR - Final: CDR	I A	ETU, FM1D
CV-2	Calibration/Validation Procedures	- Draft: 60 days prior to use - Final: 10 days prior to use <u>On-orbit Procedures</u> - Final: L-90 days	I A A	ETU, FM1D On-Orbit Procedures FM1P
CV-3	Calibration/Validation Reports Results	- Electronic Data: 3 days post-test - Test Report: 10 days post-test - Analyses: At completion + 10 days, NLT PSR	I I I	ETU, FM1P, AFM
CV-4	Radiometric Math Model (RMM)/Sensor Performance Math Model (SPMM)	- ETU: ETU Summary Review - Draft: CDR - Final: PSR	I I A	ETU, FM1P, AFM
CV-5	Calibration Data Book	- ETU: ETU Summary Review - Draft: CDR - Final: PSR	I I A	ETU, FM1P, AFM
CV-6	Spectral Response Function (SRF) Measurements	- ETU: ETU Summary Review - Final: PSR	I A	ETU, FM1P, AFM
CV-7	Intercalibration Analysis and Test Data	- Final: CDR - Update: PSR – 30 days	A A	ETU, FM1P, AFM

No.	INTEGRATION AND TEST	DUE DATE	CAT	APPLIC
IT-1	Instrument Integration, Test and Verification Plan	• Draft: PDR • Final: CDR • FM1 Update: Updated version prior to FM1 I&T based on lessons learned from the ETU	I I I	ETU, FM1D, Update: FM1P
IT-2	Test Results/Reports	• Final: 10 days after test completion	I	ETU, FM1P, AFM
IT-3	Test Procedures & As-Run Test Procedures	• Pre-Test: 10 days prior to test • As-Run: Conclusion of each test <u>Launch site-specific Test Procedures</u> • Draft: LRD – 12 months • Final: LRD – 3 months (i.e., SC PSR)	I for ETU A for Pre-Test I for As-Run I A	ETU, FM1P, AFM
IT-4	Instrument Post Delivery Support (PDS) Document	• Final: SMBA Instrument PSR minus 30 days	I	FM1P

No.	MISSION ASSURANCE	DUE DATE	CAT	APPLIC
Reserved				
MA-2	Orbital Debris Assessment Report (ODAR) and End of Mission Plan (EOMP)	- Preliminary ODAR inputs 15 days prior to PDR - Interim ODAR and Draft EOMP inputs 60 days prior to CDR - Final/updated ODAR and EOMP inputs 90 days prior to PSR	I I I	FM1D, FM1P
MA-3	System Safety Plan	- Draft: SRR - Final: PDR – 45 days - Updates 30 days prior to implementation	I I I	ETU, FM1D
MA-4	Safety Requirements Compliance Checklist	- Draft: SRR – 45 days - Final 45 days prior to PDR - Updates 45 days prior to CDR	A A A	ETU, FM1D, AFM
MA-5	Instrument Safety Assessment Report	- Draft: PDR – 30 days - Intermediate: CDR – 30 days - Final (with VTL): PSR – 45 days	A A A	FM1D, FM1P, AFM
MA-6	FMECA and Critical Items List (CIL)	- Draft: PDR – 30 days - Final: CDR – 30 days - Update: each milestone review leading up to launch	I I	FM1D, FM1P
MA-7	Master EEE Parts List	- EEE parts to the Parts Control Board prior to inclusion - Updates prior to inclusion	I I	FM1D, AFM
Reserved				
Reserved				

No.	MISSION ASSURANCE	DUE DATE	CAT	APPLIC
MA-10	Materials Identification and Usage List (MIUL)	- Draft: PDR – 30 days - Final: CDR – 30 days - Updates: 30 days after identification	I I I	FM1D, AFM
MA-11	Life Test Plan and Final Report for Lubricated Mechanisms	- Plan: PDR – 30 days - Report: 30 days after acceptance test completion	I I	ETU, FM1D, AFM
Reserved				
MA-13	End Item Acceptance Data Package	- Component/Subsystem Final: End Item Acceptance Data Package completion, or prior to consent to proceed to next level of integration, whichever is sooner. - Draft: PSR – 30 days - Final: 30 days prior to end item delivery - Updates: 30 days after identification	I (for comp/subsys) I A A	ETU, FM1D, AFM

No.	ON ORBIT	DUE DATE	CAT	APPLIC
OO-1	Instrument On-Orbit Commissioning Plan	• Final: PSR	I	FM1P
OO-2	Instrument User's Manual	• Final: PSR	I	ETU, FM1P
OO-3	Commissioning Report	• Prelim: 10 Days after each test • Test Results: PLAR • Final: PLAR	I I I	FM1P, AFM
OO-4	Command and Telemetry (C&T) Handbook and C&T Database	• Draft: PER - 30 days • Final: PSR – 6 months	I I	ETU, FM1P
OO-5	Instrument Launch and On-Orbit Operations and Contingency Procedures	• Final: PSR	I	FM1P
OO-6	Instrument Simulator Users Guide	• Draft: CDR • Final: PSR	I I	FM1D, FM1P
OO-7	Command and Telemetry Database (CTDB) Validation Plan	• Draft: PER - 30 days • Final: PSR – 30 days	I I	FM1D, FM1P

No.	MANAGEMENT	DUE DATE	CAT	APPLIC
PM-1	Monthly Project Management Reviews (PMR)	Monthly at review	I	ETU, FM1D, FM1P, AFM
PM-2	Engineering Peer Review Plan	- Initial: with Proposal - Final: 60 DACA	I I	ETU, FM1D
PM-3	Integrated Program Management Data and Analysis Report (IPMDAR)	- Initial: Draft Integrated Master Schedule with Proposal - Draft Baseline: 45 DACA - Baseline: 12th working day of the 3rd fiscal month after contract definitization - Monthly financials: by the 12th working day following the close of the prior month accounting period - Monthly IMS snapshot no later than the 5th working day of the month	I I A I I	ETU, FM1D, FM1P, AFM

No.	REVIEWS	DUE DATE	CAT	APPLIC
RE-1	Phase I Kickoff Meeting Data Package	• Final: 14 DACA of Phase I	I	ETU, FM1D
RE-2	Phase II Kickoff Meeting Data Package	• Final: 14 DACA of Phase II	I	FM1P
RE-3	System Requirements Review (SRR)/System Definition Review (SDR) Data Package	• Final: 5 days prior to review	I I	ETU, FM1D
RE-4	Preliminary Design Review (PDR) Data or Delta PDR Data Package	• Final: 5 days prior to review	I	ETU, FM1D
RE-5	Critical Design Review (CDR) or Delta CDR Data Package	• Final: 5 days prior to review	I I	ETU, FM1D
RE-6	Pre-Environmental Review (PER) or Delta PER Data Package	• Final: 5 days prior to review	I I	ETU, FM1P, AFM
RE-7	Pre-Ship Review (PSR) Data Package	• Final: 5 days prior to review	I I	ETU, FM1P, AFM
RE-8	Integrated Baseline Review (IBR)	• Final: 5 days prior to review	I I	ETU, FM1D
RE-9	Engineering Peer Review Data Packages	• Engineering Peer Review (EPR) Meeting Material : EPR – 3 days • Final: Electronic or Hard copies at review	I I	ETU, FM1D
RE-10	Manufacturing Readiness Review Package	• Final: 5 days prior to review	I I	ETU, FM1P
RE-11	Design Modification Review (DMR) Package	• Final: 5 days prior to review	I	FM1P, AFM
Reserved				
RE-13	Test Readiness Review (TRR) Data Package	• Final: 5 days prior to review	I	ETU, FM1P, AFM

No.	REVIEWS	DUE DATE	CAT	APPLIC
RE-14	ETU Test Summary Review Package	• Prelim: CDR • Final: 912 DACA (2.5 yrs after contract award)	I I	ETU
RE-15	Inputs to Mission Level Reviews	• Final: 14 days prior to review	I I I	ETU, FM1P, AFM

No.	SYSTEMS ENGINEERING	DUE DATE	CAT	APPLIC
SE-1	Engineering Change Requests	• Class I: As Generated	A	ETU, FM1P, AFM
SE-2	Trend Plan/Analysis, Reports, and Operations Log	• Initial: PDR • Update: PER • Final: PSR	I I A	ETU, FM1D, FM1P, AFM
SE-3	Engineering Drawings	• Indentured Drawing List: Make Available on Request • Individual Drawings: Make Available on Request	N/A	ETU, FM1D, FM1P, AFM
SE-4	System Performance Verification Plan and Requirement Verification Matrix (RVM)	• Initial: SRR-60 days • Final: PER - 21 days • Update: PSR	I A A	Plan: ETU & FM1D RVM: ETU, FM1P, AFM
SE-5	Instrument Performance Specification	• Initial: SRR-60 days • Final: CDR	I I	ETU, FM1D
SE-6	Analysis and Modeling			
SE-6a	Structural and Mechanical Analysis and Models	• Initial: PDR – 30 days • Final: CDR – 30 days • Correlated Model: PSR – 30 days	I A A	ETU, FM1P, AFM

No.	SYSTEMS ENGINEERING	DUE DATE	CAT	APPLIC
SE-6b	Thermal Analysis and Models	- Instrument DTM 1 & RTM 1: CDR – 60 days - Instrument DTM 2 + TVAC GSE: 90 days before PER - Correlated As-tested instrument DTM 3 + TVAC GSE: 90 days after instrument TVAC thermal balances - Correlated instrument DTM 4: PSR – 40 days Flight Cases Launch/Ascent/Deploy Science calibration maneuvers - Correlated instrument RTM 2: PSR-20 days If model deliveries per this schedule are not achievable due to the data analysis schedule, the Contractor shall coordinate with the COR to agree on acceptable delivery dates.	I I I I A	ETU, FM1P, AFM
SE-6c	Contamination Analysis Report	- Initial: CDR - Final: PER - Update: PSR – 30 days, correlated model,	I I	ETU, FM1P, AFM
SE-6d	Electromagnetic Interference (EMI)/Electromagnetic Capability (EMC) Analysis Report	- Initial: CDR - Final: PSR – 30 days	I A	ETU, FM1P, AFM
SE-6e	Hyperspectral Capability Analysis	- Initial: CDR - Final: PSR – 30 days	I A	ETU, FM1P, AFM
SE-6f	Radio Frequency Interference Detection Capability Analysis	- Initial: CDR - Final: PSR – 30 days	I A	ETU, FM1P, AFM
SE-7	Instrument Concept of Operations Document	- Initial: SRR - Final: PDR	I I	FM1D

No.	SYSTEMS ENGINEERING	DUE DATE	CAT	APPLIC
SE-8	Inputs to the Instrument to Spacecraft Interface Control Document (ICD)	- Initial: PDR - Final: CDR - Update: on spacecraft vendor change	I A A	FM1D
SE-9	Inputs to the Instrument to Spacecraft Mechanical Interface Control Document (MICD)	- Initial: PDR - Final: CDR - Update: on spacecraft vendor change	I A A	FM1D
SE-10	Field Programmable Gate Array (FPGA) Design Data Package.	- Prelim: PDR- 30 days - Final: CDR – 45 days - As generated for each FPGA	I I	ETU, FM1D
SE-11	Mass Properties Report	Final: PSR	I	ETU & FM1P, AFM
SE-12	Spare Parts Plan and List	Final: at Kickoff I	A	ETU, FM1D
SE-13	Firmware Design Data Package	- Final: CDR - Update: Firmware Review - 10 days	I I	ETU, FM1D

No.	SOFTWARE	DUE DATE	CAT	APPLIC
SW-1	Instrument Software Development Plan	Final: 30 DACA	A	ETU, FM1D
SW-2	Software Detailed Design Document	- Final: 30 DACA - Simulator Final: 90 DACA	I I	ETU, FM1D
SW-3	Flight Software Source Code and Version Description Document	- ETU, FM1, AFM Final: 30 days after acceptance test completion and with each FSW release/build - FSTB Final: FSTB TRR – 30 days - Simulator Final: Simulator Software Acceptance Review – 5 days	I I I	ETU, FM1P,
SW-4	Instrument Software Maintenance Plan	Final: 30 DACA	I	ETU, FM1D
SW-5	Instrument Data Sets	Final: PSR	I	ETU, FM1P, AFM

2 DATA ITEM DESCRIPTIONS

2.1 Calibration/Validation Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-1

2. Title:

CALIBRATION/VALIDATION AND
INTERCALIBRATION PLAN

3. Reference:

SOW 5.6.2

4. Use:

Document defines the planned calibration and validation (Cal/Val) activities to be performed by the Contractor. The scope is defined for a single ETU/FM and includes inter-calibration planning, radio frequency interference (RFI) detection and hyperspectral capabilities.

5. Preparation Information:

The Contractor shall work with the Government's Sensor Data Records Team to develop a Cal/Val Plan that describes the approach for calibrating, characterizing, and validating the spectral, spatial, radiometric, and geometric performance of the SMBA. The Calibration and Validation Plan shall address all requirements and tests related to science performance and Cal/Val, including inter-calibration, RFI detection and hyperspectral capabilities. This document shall include On-Orbit calibration and validation requirements and references to the operations documents that describe these calibration and validation procedures. The Contractor's comprehensive radiometric/geometric calibration plan shall include preliminary work at subsystem levels of assembly as well as at the system level and component level (such as PRT sensor, Reflector emissivity).

The Cal/Val Plan shall incorporate the following information at a minimum:

- A. A description of planned tests and analyses including:
 - a. What is being tested or analyzed and how it relates to instrument performance
 - b. The expected test or analysis results
 - c. Integration level for test or analysis model, i.e., part, subassembly, assembly, instrument
 - d. Environmental conditions for test, e.g., ambient, thermal-vacuum, on-orbit
 - e. Sampling methods and their statistical validity
 - f. Operational phase of testing, i.e., pre-launch, or commissioning

-
- g. Theoretical basis for the test or analysis, i.e., how the test is performed, how the data are reduced and the method rationale supported by fundamental physics principles and equations)
 - h. The resolution, precision and accuracy of the results and relation to the expected results
 - i. Test configuration, i.e., equipment, test equipment calibration, and test setup
 - j. Description of test or analysis results usage, i.e., processing algorithms that use test or analysis results or calibration parameters generated by the test or analysis
 - B. A test and analysis schedule and flow chart
 - C. How test/analysis results are made available
 - D. Government access and participation in pre-launch testing including a Government-owned standards or Government-performed Cal/Val-related characterization activities
 - E. Description of commercial-off-the-shelf (COTS) and custom analysis tools
 - F. A description of the On-Orbit Calibration capabilities of the SMBA, including their design, characterization techniques, and operational implementation. This shall include a description of any necessary calibration data that must be collected during normal operations.
 - G. Reference Standards and their calibration traceability
 - H. Support data requirements
 - I. Tests to be conducted for calibrating the radiometric response of the Instrument
 - J. Inter-calibration
 - a. Tests to be conducted for intercalibrating the radiometric response of the Instrument to the NIST traceable standard.
 - b. The method of intercalibrating proxy standards to the primary NIST traceable standard.
 - c. Reference Standards and their calibration traceability.
 - d. An analysis of the uncertainties associated with the intercalibration.

Given known issues with diffuser materials on other NASA missions being contaminated in manufacturing or in pre-launch testing, if these materials are used the Contractor shall ensure the Instrument Calibration/Validation Plan also includes a description of diffuser contamination management activities and characterizations including:

1. Vacuum bakes
2. Use of witness samples

The Committee on Earth Observation Satellites Working Group on Calibration & Validation provides the below definitions for calibration and validation. Reference <https://ceos.org/ourwork/workinggroups/wgcv/>.

- Calibration is the process of quantitatively defining a system's response to known and controlled signal inputs
- Validation, on the other hand, is the process of assessing, by independent means, the quality of the data products derived from those system outputs

Elaborating on these terms in the context of SMBA:

- Calibration inputs are electromagnetic (EM) energy from a scene at certain frequencies and the response (outputs) are the integrated energy under a channel spectral response function
- The data products to be validated (i.e., assessed for quality) are:
 - Raw Data Records (RDR)
 - Temperature Data Records (TDR)
 - Sensor Data Records (SDR)
 - Environmental Data Records (EDR)

Note: Per NS1-SMBA-REQ-0001, Section 1.4, the instrument will produce RDR files. The NOAA Ground System (NGS) will use RDR data and instrument calibration data to produce higher-level data products such as TDRs, SDRs, EDRs. The CV-1 Cal/Val Plan only needs to describe Contractor activities for validation of RDR and validation of the SMBA calibration data used as inputs for TDRs, SDRs, EDRs. The document does not need to describe the entirety of TDR, SDR, and EDR validation, as those processes are controlled by NOAA.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-2

2. Title:

CALIBRATION/VALIDATION PROCEDURES

3. Reference:

SOW 5.6.2

4. Use:

To verify that the planned Calibration, Validation, Intercalibration, RFI detection and hyperspectral procedures meet the goals of the CV-1 Cal/Val and Intercalibration Plan.

5. Preparation Information:

1. For each test in the Calibration/Validation Plan, the Calibration, Validation or Intercalibration procedure for each test shall include:
 - a. Test Objectives
 - b. Test Methods
 - c. Assumptions
 - d. Applicable Documents and Software
 - e. Associated algorithms
 - f. Required Instrument Configuration, including any differences between pre-flight and flight configurations
 - g. Mechanical and Electrical Test Equipment Configuration, including layout and interconnection of test equipment and articles including the grounding scheme. Location and identification of all measuring points on appropriate schematics and diagrams
 - h. Test Equipment Identification
 - i. Test Instrumentation
 - j. Calibration reference requirements
 - k. Safety Provisions and Cautions, including Identification of hazardous and potentially hazardous situations and operations and abort conditions
 - l. Environmental and/or other conditions to be maintained, including contamination controls
 - m. Personnel responsibilities and chain-of-command for test performance
 - n. Expected results in telemetry and associated caution and warning levels.
 - o. Program Quality Requirements
 - p. Stimuli sources and associated and associated levels
 - q. Test durations
 - r. Test configurations

- s. Sequence of events
 - t. Trending of performance characteristics during verification testing
 - u. Step-by-step instructions
 - v. Data Recording/Output Format Requirements
 - w. Data Recording Forms and Tables
 - x. Analysis techniques
 - y. Expected results
 - z. Pass/Fail Criteria
 - aa. Test Termination Procedure
2. For each Analysis in the Calibration/Validation Plan, the procedure for each Analysis shall include:
- a. Analysis Objectives, i.e., short description of the item or items being validated by this analysis
 - b. Analysis Methods including the analysis level of depth (i.e., 1st order, 2nd order, etc.)
 - c. Assumptions
 - d. Applicable Documents and Software
 - e. Associated algorithms and software tools
 - f. Required analysis mode/model configuration(s)
 - g. List of input data that will be used and related test IDs that were used to acquire the data
 - h. List of external reference sources of input data and/or model used in analysis
 - i. List of fixed analysis parameters and set points
 - j. List of adjustable analysis parameters and set points that may vary between runs.
 - k. Equipment requirements
 - l. Program Quality Requirements
 - m. Sequence of events
 - n. Step-by-step description of the data analysis procedure/algorithm
 - o. Description of any statistical sampling method being used
 - p. Data Recording/Output Format Requirements
 - q. Data Recording Forms and Tables
 - r. Expected results
 - s. Pass/Fail Criteria

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-3

2. Title:CALIBRATION/VALIDATION RESULTS

3. Reference:SOW 5.6.2

4. Use:

To provide results of individual calibration/validation, intercalibration, RFI detection, and hyperspectral tests and analyses.

5. Preparation Information:

The Contractor shall provide the Calibration/Validation Results for each calibration, validation or intercalibration test or analysis. All supporting data shall be provided in electronic format, where applicable.

The Calibration/Validation Results shall include at a minimum:

- A. Identification of article or component tested or being validated with date of test or related analysis and the relevant Test/Analysis ID and run number
- B. Performance trends during and between each planned test
- C. List of uncertainty levels for each input data in analysis or test measurements
- D. If a given test or analysis has more than one configuration state the test or analysis configuration selected.
- E. Problems or failures with tests, procedures, or analyses
- F. Anomalies and deviations from plans or procedures and their resolution/status
- G. Test or analysis results, including:
 - a. any calibration parameters to be used for on-orbit processing
 - b. comparison of results with expectations and requirements
 - c. interpretation of the results
- H. Instrument Data Sets
 - a. General Notes
 - 1. These data sets shall be provided on standard commercially available digital media.
 - 2. These data shall be provided in a format accompanied by corresponding descriptive documentation.
 - 3. For Instrument and Observatory level testing where test pass/fail status reporting is required in a timely manner during the execution of the test (for example in

thermal vacuum (TVAC) before the consent to proceed to end a plateau), quick look data sufficient to evaluate the tests pass/fail status shall be provided in near real time.

- b. For instrument-level test output these data sets shall consist of:
 - 1. Unprocessed detector sample data with associated time indexing information.
 - 2. Instrument ancillary data that would be included in the SMBA mission data stream and associated time indexing information.
 - 3. Instrument housekeeping information that would be included in the observatory telemetry stream and associated time indexing information.
- c. For observatory-level test output these data sets shall consist of:
 - 1. Unprocessed detector sample data with associated time indexing information.
 - 2. Instrument ancillary data that would be included in the SMBA mission data stream and associated time indexing information.
 - 3. Observatory ancillary data that would be included in the SMBA mission data stream and associated time indexing information. Ancillary data fields that are not available or are not meaningful in the prelaunch environment (e.g., ephemeris) may be simulated.
 - 4. Instrument housekeeping information that would be included in the observatory telemetry stream and associated time indexing information.
 - 5. Observatory housekeeping information that would be included in the observatory telemetry stream and associated time indexing information.

Characterization measurements of noise power spectral density shall be made for all continuity channels and provided in the Calibration/Validation reports.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-4

2. Title:RADIOMETRIC MATH MODEL
(RMM)/SENSOR PERFORMANCE MATH
MODEL (SPMM)3. Reference:

SOW 2.5.1, 2.5.1.6, 5.6.2

4. Use:

For modeling, evaluating, and characterizing the on-orbit end-to-end radiometric performance of the instrument.

5. Preparation Information:

- A. The RMM/SPMM shall be used to: evaluate the end-to-end radiometric performance of the Instrument (Photons to calibrated brightness temperature); conduct sensitivity analyses; determine absolute and relative calibration accuracies; identify error contributors which can be eliminated or mitigated during the design phase; identify impact of error budget trades; assess instrument performance in terms of Noise Figure (NF), Noise Equivalent Delta Temperature (NE Δ T), stability in orbit, etc., including radiometric sensitivity, calibration accuracy, nonlinearity, and noise; verify the emissivity of any on-board blackbody calibration references is > 0.99990 per PRD requirement SMBA-22.
- B. The model shall incorporate actual test and calibration data; the model shall be updated and refined during the course of the Instrument development program until it simulates instrument performance within the accuracy required by the specifications.
- C. The model shall include calibration algorithms. The estimated data shall be replaced with measured data when it is available. The post-launch calibration algorithms are used along with pre-launch calibration data to demonstrate that the absolute and relative radiometric accuracies are being met. The calibration data shall also be provided in a mutually agreed upon electronic format. Algorithms to be provided shall include:
 - 1. Radiometric Temperature Computation
 - 2. Target Temperature Computation
 - a. Physical Temperature Computation
 - b. Brightness Temperature Computation

3. Calibration Accuracy/Uncertainty Measurement
4. Noise Power Spectral Density (NPSD) and NE Δ T Measurements
5. non-Linearity
6. Dynamic Range
7. Gain Stability ($\Delta G/G$)
8. Hot and Cold Calibration References
 - a. Hot Target calibration reference uncertainty and corrections
 - b. Cold Target calibration reference uncertainty and corrections

The algorithm delivery shall consist of:

- a. An overview of each algorithm including the objective or purpose of the algorithm
 - b. The rationale for each algorithm and identification of any specifications it supports
 - c. A description of each algorithm's input parameters, mathematical equations, and resulting outputs, including the background and justification for their use, and a description of any external ancillary data (i.e., from the spacecraft) used by the algorithm
 - d. A mathematical description of each algorithm
 - e. An implementation of the algorithm in source code and identification of the Operating System that the source code was built on
 - f. An error analysis showing the expected accuracy of the algorithm results
 - g. A description of the methods used to validate each algorithm and description of the validation results
 - h. A list of the instrument data components used by each algorithm
 - i. A list of the calibration parameters used by each algorithm
 - j. A list of references
- D. The SPM shall also propagate uncertainties in model input parameters to uncertainties in model output parameters, including error correlation effects. The basis for uncertainties in SPM inputs shall be documented. The delivered SPM shall include full documentation of the model, including the error propagation methodology, any software code/scripts to run the model, and instructions for Government customer use of the model.
- E. System performance predictions shall include uncertainty estimates for calibrated radiance, reflectance, and equivalent blackbody temperatures based on propagation of uncertainties in SDR algorithm look-up table values and other SDR algorithm parameters.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-5

2. Title:

CALIBRATION DATA BOOK

3. Reference:

SOW 5.6.2

4. Use:

For demonstrating specification compliance and for full interpretation of the orbital data.

5. Preparation Information:

The calibration data book is to be prepared and processed for public release prior to delivery to NASA.

The calibration data book, which is to be delivered at the end of pre-launch instrument calibration and intercalibration tests and as part of Pre-Ship Review (PSR) Data Package, are to be a record of all pertinent test, calibration and intercalibration data, including all raw data from which graphs, curves, etc., are prepared, and whose primary functions include 1) providing NOAA calibration parameters for RDR, TDR, and SDR generation and 2) providing users' info needed to process data.

The calibration data book shall be updated after commissioning Cal/Val activities.

The Cal Data Book will provide calibration parameters needed by NOAA (or NGS) to produce TDRs and SDRs.

A summary section shall be prepared, which contains functional equations and charts pertinent to the data contained therein. The summary shall depict, as a minimum, all information pertinent to the performance requirements, the final system level bench test data, the final calibration and intercalibration of the instrument and every instrument subsystem as determined from all tests and calibrations performed prior to delivery of the instrument to the Government, and a set of trend charts of all critical parameters.

The calibration book data shall be submitted in accordance with Table 1 of this document.

Data shall be provided for each tested instrument temperature.

The results of all tests and calibrations shall be summarized by the contractor in the calibration log book(s).

The following information shall be derived from the calibration data.

- (a) Temperature sensitivity ($NE\Delta T$) of each channel
- (b) Standard deviation of individual calibration points from the best-fit calibration curves for each channel and each calibration
- (c) Calibration of in-flight calibration blackbodies in terms of effective brightness temperature versus temperature monitor bit outputs.
- (d) Calibration curves of all temperature sensors.
- (e) Calibration curves of all voltage monitors.
- (f) Temperature stability data for the instrument.
- (g) Beam profile in the far field for each channel

Spectral Response Function (SRF) frequency data shall be in the radiofrequency (RF). This ensures that data is delivered at RF frequencies, not the intermediate frequency (IF).

SRF data shall be delivered in digital format (such as .csv or MATLAB) that includes numerical values for gain vs. frequency and graphical plots of the SRFs. The data format is needed so that Government can process and analyze SRF data.

SRFs, and associated text describing the SRFs, shall be prepared for public release and included in the Calibration Data Book. SRFs are planned for public release to be made available to data users.

Pre-launch calibration parameters required for data processing shall be delivered in the Calibration Data Book.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-6

2. Title:Spectral Response Function (SRF) Measurements

3. Reference:PRD Sec 4.9; SOW 5.1, 5.6.2

4. Use:To provide the SRF data of each channel.

5. Preparation Information:

How the data was collected, the raw measurement data, analysis of data and any corrections to the data needed prior to use by data users, final data set after all corrections have been applied, text to be associated with data for use in CV-5.

SRF frequency data shall be in the radiofrequency (RF).

SRF data shall characterize SRF changes over the operational lifetime of the instrument.

SRF data shall be delivered in digital format (such as .csv or MATLAB) that includes numerical values for gain vs. frequency and graphical plots of the SRFs.

SRFs, and associated text describing the SRFs, shall be prepared for public release and included in CV-5 Calibration Data Book.

DATA ITEM DESCRIPTION

1. CDRL No.:

CV-7

2. Title:INTERCALIBRATION ANALYSIS AND TEST
DATA3. Reference:

SOW 2.5.1.9, 5.6.2

4. Use:

To provide results from the intercalibration activities of the SMBA units for Government review. CV-7 documents the measurements and analyses from the tests performed with the SMBA units that are needed to complete intercalibration across those units. Of key importance is data to enable quantification of the main sources of uncertainty, and any applicable limitations on the test data, plus key assumptions made during the analysis.

5. Preparation Information:

Calibration, intercalibration, and data processing algorithms used shall be provided with the analysis of the test data. Test data shall include how the data was collected, the raw measurement data, analysis of data and any corrections to the data needed prior to use by data users, final data set after all corrections have been applied, text to be associated with data for use in CV-3.

2.2 Integration and Test Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-1

2. Title:

INSTRUMENT INTEGRATION, TEST AND
VERIFICATION PLAN

3. Reference:

SOW 4.2.4, 4.3.1, 4.3.2, 5, 5.6.1

4. Use:

To provide information on the integration, testing and verification of the instrument, including the ETU and flight software. To provide an overview of the entire integration and test (I&T) flow and its relation to performance verification. To describe the methods, policies, and organization of the ETU and flight unit test program. To define the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing. To define the flight software test (including acceptance test) plan including for new software capabilities for SMBA such as RFI detection and hyperspectral.

This DID includes all the Contractor generated documents necessary to perform the integration and testing of the instrument (e.g., the Environmental Verification Plan (EVP), the Environmental Test Matrix, ICD/PRD verification, etc.). The EVP documents the Contractor's approach for environmental qualification and acceptance tests. The Environmental Test Matrix summarizes the tests performed. This DID also satisfies the environmental verification requirements of the launch services provider.

5. Preparation Information:

The Contractor shall provide an Instrument I&T Plan which describes the series of activities required to integrate the various instrument components into the final flight configuration. The Contractor shall prepare a test plan describing the methods, policies, and organization of the flight test program. The plan shall also define the test requirements and the tests to be performed, instrumentation and locations, the tests methods, stimuli/measurements for acceptance criteria, and a time sequence for the testing.

The I&T plan shall define and describe the series of activities required to test the flight software including acceptance testing.

The I&T Plan shall cover the ETU, Flight Model 1 and all Flight Models and clearly identify any differences in approach or activities between the ETU and each of the subsequent flight models.

The I&T Plan shall cover limitations in the instrument verification program which preclude the verification by test of any system requirement shall be documented. Examples of limitations in the ability to demonstrate requirements include:

- Inability to deploy hardware in a 1-g environment.
- Facility limitations which do not allow testing at system level of assembly.
- Inability to perform certain tests because of contamination control requirements.
- Inability to perform powered-on testing because of voltage breakdown concerns.

Alternative tests and analyses shall be evaluated and implemented as appropriate, and an assessment of program risk shall be included in the System Performance Verification Plan.

At a minimum, the plan shall contain the following information:

- (1) Deliverables to Instrument Integration and Test
- (2) Flow of instrument Integration and Test sequence, including identification of calibration tests and characterizations. Identify instrument level requirements to be verified at each stage, cross referenced to the System Performance Verification Plan (SPVP). Identify dependencies within the I&T sequence.
- (3) Description of each instrument activity or test including the general test configuration
- (4) Description of instrument procedures required to support the activity or test
- (5) Description of simulated data/simulator requirements
- (6) Description of instrument special handling requirements to support the activity or test
- (7) Description of instrument ground support equipment requirements to support the activity or test
- (8) Description of instrument safety requirements to support the activity or test
- (9) Description of instrument personnel required to support the activity or test
- (10) Requirement to take photographs during the integration process for reference
- (11) Description of the process to be used for handling anomalies as they occur during I&T
- (12) Requirement of log books to be maintained throughout the I&T process

Proposed tailoring of tests shall include a comparison (gap analysis) of the planned test program to the ICD/PRD. Note: the comparison to ICD/PRD will be used by the Government team to support future reviews with Independent Review Teams. Justification shall be provided for any requested deviations to ICD/PRD.

This CDRL shall have the following as an appendix:

As an appendix to the instrument test and verification plan, a test matrix shall be prepared that summarizes all tests that will be performed on each component and each subsystem. The purpose is to provide a ready reference to the contents of the test program in order to prevent the deletion of a portion thereof without an alternative means of accomplishing the objectives; it has

the additional purpose of ensuring that all flight hardware has been subjected to environmental exposures that are sufficient to demonstrate acceptable workmanship. In addition, the matrix shall provide traceability of the qualification heritage of hardware. All flight hardware, spares, and prototypes (when appropriate) shall be included in the matrix. Details of each test shall be provided (e.g., number of thermal cycles, temperature extremes, vibration levels). It shall also relate the design environments to the test environments and to the anticipated mission environments. The matrix shall be prepared in conjunction with the initial environmental verification plan and shall be updated as changes occur.

A complementary matrix shall be included showing the tests that have been performed on each component or subsystem (or applicable level of assembly). This should include tests performed on prototypes or engineering units used in the qualification program, and should indicate test results (pass/fail or malfunctions).

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-2

2. Title:

TEST RESULTS/REPORTS

3. Reference:

SOW 4.2.4, 5.6.6, 5.7.2, 6.4

4. Use:

Report the results of all tests identified in the Test Plans, including test procedures used, test results, configuration status of all items under test, all measured data, a summary of observed results, and a discussion of compliance.

5. Preparation Information:

Test reports shall be delivered for all major environmental tests (TVAC, Vibrations, EMI/EMC) and performance verification tests (including the CPT). For all other tests, in lieu of reports, test results can be delivered in contractor format. The results from all tests, including Flight Software tests, that are used for requirements verification shall be provided.

The following shall be included in test reports at a minimum:

- a. Test identification and hardware configuration (and software version number if applicable) – for specific test
- b. Facility description
- c. Reference – applicable test plan, test procedures, and test requirements, test log including the dates of the testing, photographs of test setup, any malfunction reports written during the test
- d. Test results, to include:
 1. Identification of test results which confirmed the expected results as specified in the test plan/procedures or for which variations between actual and expected results were within specified tolerance. For the latter case, actual test results shall be shown.
 2. Identification of test results which differ from expected results beyond expected or acceptable limits.
 3. Identification of any planned test objective or requirement for which actual results were not obtained. Reasons for not meeting the objective/requirement shall be stated.
 4. Identification of any false or aberrant results noted during the test or subsequent analyses.
- e. Recommendations for subsequent actions shall be stated, based on the test results.

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-3

2. Title:DETAILED TEST PROCEDURES AND AS-RUN
TEST PROCEDURES3. Reference:

SOW 1.5.1.9, 4.2.4, 4.3.1, 4.3.2, 5.6.5, 6.6, 6.6.2.3

4. Use:

To provide information on how the Instrument (system-level, hardware, software, etc.) and associated hardware such as cables and Ground Support Equipment (GSE)/System Test Equipment (STE) will be tested and how their performance will be verified.

For As-Run Procedures, to provide an archive of test procedures used during system development to aid in on-orbit anomaly resolution.

5. Preparation Information:

The Contractor shall provide detailed test procedures to be used during Instrument Level Testing (system-level, hardware, software, etc.). Test procedures shall provide information sufficient to understand the purpose and methodology of all tests, and to provide the required environmental and configuration controls necessary for successful completion of the test. These procedures shall be prepared for each test activity defined in the Performance Verification Plan (CDRL SE-4) and shall cover all Instrument test operations, interfaces, and Instrument performance requirements (i.e., electrical, structural and mechanical, EMC), and shall cover specialized tests such as mechanical function and deployments, environmental exposure tests (i.e., vacuum, vibration), Instrument calibration, GSE/STE calibration and checkout, and pre-launch end-to-end tests. If tests are conducted in conjunction as part of a “group” test, for example, Limited Performance Test (LPT) or Comprehensive Performance Test (CPT), one test procedure may encompass this group. At a minimum, the procedures shall contain the following information:

- a. Test Objectives including hardware and/or software to be tested
- b. Test Methods and Sequence, including Test methods documented within a test procedure prior to start of test
- c. Applicable Documents and Software
- d. Required Instrument configuration, including any differences from flight configuration
- e. Test Equipment Configuration, including layout and interconnection of test equipment and articles including the grounding scheme. Location and identification of all measuring points on appropriate schematics and diagrams

-
- f. Test Equipment and Facility Identification (Test equipment & tooling lists including type, range, and accuracy for each item)
 - g. Test Instrumentation
 - h. A confirmation that each piece of test equipment is within its calibration period.
 - i. Safety Provisions and Cautions, including Identification of hazardous and potentially hazardous situations and operations and abort conditions
 - j. Environmental and/or other conditions to be maintained, including contamination controls
 - k. Responsibilities and chain-of-command for test performance
 - l. Expected results in telemetry and associated caution and warning levels.
 - m. Data Recording Requirements
 - n. Data Recording Forms and Tables
 - o. Accept/Reject Criteria
 - p. Note any test phases and profiles
 - q. List the requirements for the test procedure and test report development
 - r. Description of any necessary functional operations required during the test (i.e., a CPT performed at hot and cold during thermal vacuum testing)
 - s. Test parameters/specific data to be measured
 - t. Test limits and tolerances
 - u. Inspections points as applicable
 - v. Consent to Proceed (CTP)/Consent to Break (CTB) points associated with the test (as applicable)

As-Run Test Procedures shall be all as-run, red-lined, Contractor-format test procedures used during instrument development, integration, and test at instrument and at observatory levels. With the as-run procedures, include all deviations from the test procedures with a rationale for each deviation. The Contractor shall provide Automated Test Scripts.

DATA ITEM DESCRIPTION

1. CDRL No.:

IT-4

2. Title:Instrument Delivery and Post Delivery Support
(PDS) Document3. Reference:

SOW 6.2, 6.6

4. Use:

This document is primarily intended to define the required instrument operations from receipt at the spacecraft contractor's facility through encapsulation. This document will include the details pertaining to equipment, personnel, testing, safety, and contamination providing the baseline integration plan, description, and procedure for conducting post-delivery activities from instrument delivery at the Spacecraft Contractor to encapsulation for SMBA. This document will be used to guide the spacecraft and instrument team to complete all operations and activities the instrument requires to be ready for flight. This document will be used by the Government and Spacecraft Contractor to prepare a successful satellite integration and test campaign.

This document is also intended to cover packaging, handling and transportation of the SMBA instrument to the spacecraft vendor.

5. Preparation Information:

The Contractor shall provide the stated information below in sufficient detail to meet the intended uses of this document as stated above.

Equipment details shall include all equipment required for every test event at the spacecraft contractor's facility and at the launch site. The information should be delineated by type of equipment, where and when the equipment is used, if it is temporary, what events the equipment is used for, operating restrictions for the equipment, equipment safety concerns, and equipment cleaning/maintenance required.

Personnel details shall include all personnel required for all types of test events and all operations along with their roles and responsibilities. Each test event or operational should clearly describe what instrument personnel are required, what their roles are, and whether they are responsible for the operation or whether they are there to support the spacecraft contractor personnel.

Testing details shall include a mapping of all instrument test activities mapped to each satellite level test event from Bench Acceptance Test (BAT) to launch site testing. The testing details should also include any required spacecraft configurations, any personnel required for the activity, all equipment or GSE needed for the activity, all applicable constraints, and any safety considerations.

Safety details shall include what type of safety concern the operation presents, what personnel are required, constraints to be referenced, and if any special equipment is needed.

Contamination details shall define the cleaning and operational processes throughout all events at the spacecraft contractor's facility, in transport, and at the launch site. These details should include measurements needed, any purge plans, any cleanings required, any cleaning supplies required along with which organization is supplying them, and any spacecraft/instrument configuration requirements (bags/covers/etc.). It shall identify contamination sensitivities, sources, and concerns.

The contractor shall provide the instrument scripts or inputs to spacecraft (SC) scripts necessary to run the instrument after delivery.

Plans, requirements, and procedures for maintaining and monitoring the instruments while in storage at the spacecraft contractor's facilities, and the actions to be taken in case of anomalies.

Plans, requirements, and procedures for packaging, handling, and transporting the instruments to the spacecraft contractor's facilities, and the actions to be taken in case of anomalies.

2.3 Mission Assurance Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-2

2. Title:

Orbital Debris Assessment Report (ODAR) End of Mission Plan (EOMP)

3. Reference:

SOW 3.5
NASA-STD-8719.14 Process for Limiting Orbital Debris

4. Use:

Provide information for development of the ODAR and the EOMP deliveries per the content defined in NASA-STD 8719.14

5. Preparation Information:

NASA-STD-8719.14 Process for Limiting Orbital Debris Appendix A (ODAR) and Appendix B (EOMP) provide details on what information is required for the Project office to complete these analyses.

NOTE: Orbital Debris Assessment Software is available for download from Johnson Space Center at <https://orbitaldebris.jsc.nasa.gov/mitigation/debris-assessment-software.html>.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-3

2. Title:

System Safety Plan

3. Reference:

SOW 3.6.1

NASA-STD-8719.24 (with Annex), NASA Launch Vehicle Payload Safety Requirements

4. Use:

The contractor shall prepare a SSP that describes the development and implementation of a system safety program that complies with the requirements of NPR 8715.7, the launch service provider, and launch range safety.

5. Preparation Information:

The contractor shall

- a) Define the roles and responsibilities of personnel.
- b) Define the required documentation, applicable requirements documents, and completion schedules for analyses, reviews, and safety packages.
- c) Address support for Safety Reviews, Safety Working Group Meetings and Technical Interchange Meetings.
- d) Provide for early identification and control of hazards to personnel, facilities, support equipment, and the flight system during product development, including design, fabrication, test, transportation, and ground activities.
- e) Address compliance with the launch range safety requirements.
- f) Include a safety review process that meets the requirements of NPR 8715.7 Payloads Safety Program.
- g) Address compliance with safety requirements imposed by NASA and Occupational Safety and Health Administration (OSHA), address design and operational safety requirements, and address contractually imposed mission unique obligations.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-4

2. Title:

Safety Requirements Compliance Checklist

3. Reference:

SOW 3.6.4

NASA-STD-8719.24 (with Annex), NASA Launch Vehicle Payload Safety Requirements

4. Use:

The contractor shall prepare a compliance checklist of all design, test, analysis, and data submittal requirements.

5. Preparation Information:

The following shall be included:

- a) Criteria and requirement.
- b) System
- c) Indication of compliance, noncompliance, or not applicable
- d) Rationale for indications other than compliant
- e) Resolution
- f) Reference
- g) Copies of Range Safety and NASA approved non-compliances, including waivers and equivalent levels of safety certifications.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-5

2. Title:

Instrument Safety Assessment Report

3. Reference:

SOW 3.6.5

4. Use:

The ISAR will identify safety features of the hardware, software, and system design as well as procedural, hardware, and software related hazards that may be present in the instrument. This includes specific procedural controls and precautions that should be followed.

5. Preparation Information:

The ISAR will include the following information:

- a) The safety criteria and methodology used to classify and rank hazards, including assumptions upon which the criteria or methodologies were based or derived.
- b) The results of hazard analyses and tests used to identify hazards in the system including:
 - a. Those hazards that still have a residual risk and the actions that have been taken to reduce the associated risk to a level contractually specified as acceptable.
 - b. Results of tests conducted to validate safety criteria, requirements, and analyses.
 - c. Hazard reports documenting the results of the hazard analyses to include a list of all significant hazards along with specific safety recommendations or precautions required to ensure safety of personnel, property, or the environment. NOTE: Identify whether or not the risks may be expected under normal or abnormal operating conditions.
 - d. The hazardous procedures and associated safeguards that will be used for integration and test activities and pre-launch activities.

The contractor will submit with the Final ISAR a signed statement that:

- a) Hazards have been closed out through elimination or through risk mitigations that are contractually specified as acceptable.
- b) Identifies hazards that have not been eliminated or controlled at the time of submission.
- c) The instrument is ready to test, operate, or proceed to the next phase.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-6

2. Title:

FMECA and Critical Items List

3. Reference:

SOW 3.7.3

4. Use:

The FMECA Report shall explain the approach of the analysis, methodologies, assumptions, results, conclusions, and recommendations.

5. Preparation Information:

- 1) 1) The FMECA shall provide a ranking of items to determine criticality.
- 2) Appropriate retention rationale shall be stated. Rationale may include design features, historical performance, acceptance testing, manufacturing product assurance, corrective action recommendation/elimination of undesirable failure modes, proper design controls, and failure detection methods. The rationale also will contain data that describes operational constraints caused by occurrence of the failure and any measures that can be taken to restore the function on orbit where known.
- 3) The FMECA shall provide a single point failure (SPF) list for failure modes.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-7

2. Title:

Master EEE Parts List

3. Reference:

SOW 3.10.4

4. Use:

The contractor shall maintain the Master EEE Parts List Information in a searchable electronic format. The GSFC Project Parts Engineer shall have access to the list.

5. Preparation Information:

The contractor shall generate and maintain a Master Parts List with the minimum information listed below for the various stages throughout the project's lifecycle:

- a) For EEE parts that are identified for potential flight use, the list shall include the following information:
 - 1. Flight component identity to the circuit board level
 - 2. Complete part number (i.e., Defense Supply Center Columbus part number, Specification Control Drawing part number, with all suffixes)
 - 3. Manufacturer's Generic Part number
 - 4. Manufacturer (not distributor)
 - 5. Part Description (please include meaningful detail)
 - 6. Federal Supply Class
 - 7. Procurement Specification
 - 8. Comments and clarifications, as appropriate
 - 9. Estimated quantity required (for procurement forecasting)
- b) For EEE parts that are approved for flight use the list shall include the following information:
 - 1. Procurement Part Number
 - 2. Flight Part Number (if different from the procurement part number)
 - 3. Package Style/Designation
 - 4. PCB Status
 - 5. PCB Approval Date
 - 6. PCB Required Testing/Evaluations
 - 7. Total Ionizing Dose (TID) rating

8. Single Event Effects (SEE) rating
- c) For EEE parts that are approved for as-designed use the list shall include the following information:
1. Assembly Name/Number
 2. Next Level of Assembly
 3. Need Quantity
 4. Reference Designator(s)
 5. Item number (if applicable)
- d) For EEE parts in flight hardware the list shall include the following information:
1. Assembly serial number
 2. Item revision
 3. Next Level of Assembly serial number
 4. Lot/Date/Batch/Heat/Manufacturing Code, as applicable
 5. Manufacturer's Cage Code (specific plant location when relevant)
 6. Distributor/supplier, if applicable
 7. Part number
 8. Part serial number (if applicable)

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-10

2. Title:

Materials Identification and Usage List (MIUL)

3. Reference:

SOW 3.11.2

4. Use:

Provide a material list for materials used on the instrument.

5. Preparation Information:

The MIUL documentation approach, per NASA-STD-6016C, Section 4.1.5, shall be defined in the MPSCIP.

Additions and Prescribed Changes to NASA-STD-6016C MIUL requirements:

- a) The MIUL shall identify all unique materials and processes used in the construction of the deliverable hardware.
- b) The contractor may use GSFC forms or the contractor's equivalent forms in lieu of the MAPTIS format.
- c) The MIUL shall include Thermal Vacuum Stability data, when appropriate, per NASA-STD-6016C, Section 4.2.3.6.
- d) The MIUL shall include the vacuum bake-out schedule for materials that fail outgassing requirements or provide a separate MUA.
- e) The contractor may use the GSFC outgassing database (URL <http://outgassing.nasa.gov>) in addition to MAPTIS (URL <http://maptis.nasa.gov>).
- f) The MIUL shall document, when appropriate, compliance to MSFC-STD-3029A
- g) The MIUL shall clearly identify all materials that do not meet the requirements of NASA-STD-6016C.
- h) The MIUL shall include soldering flux and solvents used for cleaning flight electronic assemblies, other than isopropyl alcohol or deionized water.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-11

2. Title:

Life Test Plan and Final Report for Lubricated Mechanisms

3. Reference:

SOW 3.7.4

NASA-STD-6016C Standard Materials and Processes Requirement for Spacecraft

NASA-TM-86556 Lubrication Handbook for the Space Industry (Part A: Solid Lubricants, Part B: Liquid Lubricants)

NASA/CR-2005-213424 Lubrication for Space Applications

4. Use:

The contractor shall prepare and implement a Life Test Plan to qualify all lubricated mechanisms either by life testing or heritage with an identical mechanism used in an identical application. The contractor shall perform a lubricant loss analysis for all mechanisms to show that the design meets a 10X margin.

The contractor shall prepare a life test final report at the end of each life test.

5. Preparation Information:

The plan shall contain:

- a. Table of Contents.
- b. Description of lubricated mechanisms, performance functions, summary of subsystem specification, and life requirements.
- c. Heritage of identical mechanisms and descriptions of identical applications.
- d. Design, drawings, and lubrication system used by the mechanism.
- e. Test plan, including vacuum, temperature, and vibration test environmental conditions.
- f. Criteria for a successful test.

NASA-STD-6016C, Section 4.2.3.4, NASA-TM-86556, and NASA/CR-2005-213424 should be used as guides when developing the plan.

Alternatively, the contractor shall provide detailed justification for NOT conducting life tests and/or lubricant loss analyses.

The life test final report shall be prepared in the contractor's format.

DATA ITEM DESCRIPTION

1. CDRL No.:

MA-13

2. Title:

End Item Acceptance Data Package

3. Reference:

SOW 1.1.4.4, 3.14, 6.1

4. Use:

Provides a data package to include all paperwork necessary to assess the quality of the end item.

5. Preparation Information:

The contractor prepares the EIADP as part of design development and implementation such that it is completed prior to delivery. The following items shall be included (as specified by requirement):

- a) The deliverable item name, serial number, part number, and classification status (e.g., flight, non-flight, ground support, etc.).
- b) Appropriate approval signatures (e.g., contractor's quality representative, product design lead, Government Representative, etc.).
- c) List of shortages or open items at the time of acceptance with supporting rationale.
- d) Supplier list
- e) As-built serialization.
- f) As-built configuration.
- g) In-process Work Orders (available for review at Contractor's – not a deliverable).
- h) Final assembly and test Work Order.
- i) Major MRB reports with a minimum:
 1. Part number, serial number, problem description, disposition, root causes, and corrective actions.
- j) Major anomaly reports with a minimum:
 1. Part number, serial number, problem description, disposition, root causes, and corrective actions.
- k) Acceptance testing procedures and report(s), including environmental testing.
- l) Quality Metric Trend data
- m) AS- built Master EEE parts list.
- n) As-built materials identification and usage list.
- o) Chronological history, including:
 1. Total operating hours and failure-free hours of operation.

2. Total number of mechanical cycles and remaining cycle life.
 3. Limited life items, including data regarding the life used and remaining.
 4. As-built final assembly drawings.
 5. Any PCB coupon results.
 6. Photographic documentation of hardware (pre- and post-conformal coating for printed wiring assemblies, box or unit, subsystem, system, harness, structure, etc.).
- p) List or table of all waivers.
- q) Certificate of Compliance which is signed by management.

2.4 On Orbit Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-1

2. Title:

INSTRUMENT ON-ORBIT COMMISSIONING
PLAN

3. Reference:

SOW 6.4.1.1

4. Use:

To define how the Instrument will be activated, calibrated, and tested (functionally and performance); and how its on-orbit performance will be verified. The Plan should include test activities, test equipment and plans for inclusion of the Government test team.

5. Preparation Information:

This CDRL item may be combined with CDRL OO-5.

The On Orbit Commissioning Plan shall be the governing document for initialization and validation of the Instrument during the pre-operational check-out period. The plan shall include:

- a. A summary of the initialization and verification methodology
- b. A matrix or list of any Instrument requirements, with associated acceptance criteria, to be verified on-orbit which is cross-referenced to the appropriate Instrument On-Orbit Test Procedures or Calibration/Validation Procedures
- d. The schedule of initialization and verification activities, including start times and durations
- e. Procedure numbers of the Instrument On-Orbit Test Procedures to be used during initialization and verification
- f. Constraints to operations

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-2

2. Title:

INSTRUMENT USERS MANUAL

3. Reference:

SOW 4.2.5, 6.4.1.1, 6.4.1.2,

4. Use:

The report shall be a self-contained document in that a reader not familiar with the Instrument can obtain a reasonably complete understanding of the Instrument without recourse to another document or drawing. The document is not meant to be an engineering working document but a reference document for SMBA Instrument data users including: Observatory operators, Government personnel, scientists, Observatory contractor personnel, and the general public.

5. Preparation Information:

The Instrument User's Manual shall provide a non-proprietary description of the system, subsystems, functions, and operations (with emphasis on how the system operates), with illustrations, block diagrams and circuitry descriptions. The Instrument to spacecraft interface shall be described. Physical details (mechanical, electrical, thermal, optical train & detectors, etc.), information on the software design, interfaces, instrument modes and states, commands and telemetry, and performance characteristics shall be provided by inclusion or by reference. The report shall be a self-contained document in that a reader not familiar with the Instrument can obtain a reasonably complete understanding of the Instrument and its operation without recourse to another document or drawing. The document is not meant to be an engineering working document but a reference document for Government personnel, scientists, Observatory contractor personnel, and the general public of SMBA Instrument data users (including international users).

The SMBA Instrument User's Manual shall characterize Instrument performance with respect to: relative spectral response; radiometric accuracy, sensitivity, and stability; and line-of-sight accuracy. Each Manual shall provide pre-flight test results characterizing performance and a flight performance evaluation with updates at major milestones. The document shall describe the Instrument modes of operation and the equipment, methods, accuracies, and concepts of operation for in-flight calibration of radiometric response and line-of-sight. The document shall include a description of the data format and packet structure.

The User's Manual shall stand alone. There shall be no references to training packages or other materials (e.g., internal vendor reports/memos) that will not be readily available. Note that what

is ‘readily available’ is not the same at beginning of life (BOL) as at end of life (EOL), and the On-Orbit User’s Manual must apply to EOL.

In general, if there is any doubt whether an operator may need to know the information, the Contractor should include it.

Constraints shall be discussed as appropriate and in context; e.g., if critical commands must be enabled prior to sending, this should be discussed in the command and data handling (C&DH) section.

The User’s Manual shall cover the life of the instrument from launch pad through decommissioning.

Note the following is not intended to be a table of contents. The information below does not have to be in the order or groupings given here, but are provided to ensure expected content is included in the deliverable.

Information should include, but is not limited to, the items below:

- Sensor overview including science mission
 - o Channel Wavelengths/Frequencies and Resolutions
 - o Scan pattern
 - o Field of View (FOV)/footprint on the Earth
 - o Scan period
 - o Scan patterns
 - o A description of the hyperspectral capabilities
 - o For hyperspectral include:
 - a. Identification of which bands and frequency ranges have hyperspectral capability
 - b. The number of channels for each hyperspectral band
 - c. The start and stop frequencies, the center frequency and bandwidth of each hyperspectral channel
 - d. Performance trades and studies used to select the hyperspectral configuration, including rationale on the selected configuration
 - e. The Spectral Response Function (SRF) of each channel, including a description of which passbands are combined to form the channel (a reference to CV-10 SRFs is acceptable)
 - f. The raw data used to derive the SRF along with the frequencies associated
 - g. The description of the SRF measurement test that provides details about the test procedure (set up and instrumentation), the environmental conditions, the data generated, the data analysis and the results obtained
 - h. A description of the algorithms that combine the channels including rationale; limitations and specifications; mathematical description; a performance analysis; and assumptions such as the data sets used for simulating the environmental conditions

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- i. The center frequency stability, calibration accuracy, and NEdT of each channel
 - j. Flexibility in the frequency plan, including capabilities to update the frequencies pre- and post-launch
 - k. Power consumption impacts
 - l. Thermal impacts
 - m. Data rate impacts on downlink
 - n. Data storage impacts
 - o. Description of the signal flow and digitization, including Fast Fourier Transforms, binning, windowing, and similar functions as applicable
 - p. A description of the key contributing hardware elements and their functions
 - q. A description of the software
 - r. Digitization errors, including how closely a digitized signal matches the analog signal
 - o A description of the RFI capabilities
 - o For RFI include:
 - a. The Instrument bands that have RFI capability
 - b. The number of channels for each band
 - c. The bandwidth of each channel
 - d. The Radio Frequency Interference (RFI) detection algorithms chosen for each band
 - e. The implementation of the RFI detection algorithms
 - f. An overview of each algorithm including the objective or purpose of the algorithm
 - g. The rationale for each algorithm and identification of any specifications and limitations
 - h. A description of each algorithm's inputs and outputs including any external ancillary data (i.e., thresholds) used by the algorithm
 - i. A mathematical description of each algorithm
 - j. An implementation of the algorithm in source code and identification of the Operating System that the source code was built on.
 - k. A performance analysis of the algorithm that provides metrics on the algorithm including its detection sensitivity, its probability of detection and its false alarm rate as function of RFI characteristics
 - l. A description of the methods used to validate each algorithm and description of the validation results
 - Subsystem/Component Overview (The purpose of this section is to inform the reader of overall subsystem information; however it should cover in detail those aspects that are monitored or changeable by the operator.)
 - o Mechanical
 - Include diagrams of motors and information about bearings and lubricants, because electro-mechanical devices are notorious troublemakers
 - o Electrical
 - Nominal operating power ranges
 - o Thermal

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- Locations of all heaters and their relationship (location relative) to thermistors
 - Heater set points
 - Nominal operating temperature ranges
 - o Optical/RF/energy path through the system
 - Detector layout (if appropriate)
 - Calibration targets/sources
 - o Flight Software
 - Software summary, including: application, inventory, environment, organization, overview of operation, contingencies, alternate states, and modes of operation.
 - Functional description
 - Simplified flowchart
 - Fault Protection
 - What it monitors
 - What it does in response to ‘violations’ of monitored information
 - Description and discussion of uploadable and downloadable tables including how to know when an update is needed
 - Fault codes (or where to find)
 - Response to fault codes (or where to find)
 - Sensor Modes
 - o Description of and how to recognize each mode
 - o Allowable Transitions
 - o Recommended Transitions
 - o Indicate what transitions can occur autonomously versus by command
 - o Table of major components and their status (e.g., ON/OFF) for each mode
 - o Table of telemetry available in each mode
 - o Activities allowable in each mode (e.g., loads and/or dumps)
 - Flight model-specific signatures
 - o ‘Features’ of the specific flight model being documented
 - o Differences with existing/operating sensors should be explicitly pointed out
 - Interfaces
 - o Sensor Internal
 - o Spacecraft External
 - Point-to-point telemetry and commands
 - Survival heater locations, set points and relationship to thermistors (if not covered under sensor thermal subsystem)
 - o Databus interaction
 - o Timing, clocks, and synchronization
 - o Average and peak data rate
 - o Average and peak power consumption
 - Command and Data Handling
 - o APIDs
 - o Telemetry Rates
 - o Packet descriptions
 - o Description on on-board stored commanding and macro capabilities

- o Critical commands
 - o Database
 - Flight Telemetry Limits ‘philosophy’: i.e., How did the Contractor ‘choose’ the flight limits? E.g., Did the Contractor take the ‘tested to’ maximum and add margin? Did the Contractor take the ‘Qual’ limits and add margin? If a limits philosophy requirement is defined elsewhere, it supersedes this guidance.
- Science
 - o Algorithms
 - o On-board data processing
- Overview of Recommended Operating Procedures (ROP)
 - o Purpose and short summary of functionality
- Normal Operations
 - o On-Orbit Activation Sequence (could be a ROP or series of ROPs) including timing as determined from testing, e.g., cool-down is expected to take x minutes or x hours or x days
 - o Cal/Val Activities
 - o On-Orbit Maintenance Activities
 - Constraints/configurations for Maneuvers DELTA-V and DELTA-I maneuvers
 - o Trending Recommendations (health and safety and science performance parameters)
 - Nominal values
 - Frequency of trending
 - Limits
 - o Special Activities (e.g., Cal/Val maneuvers, mid-mission out-gassing)
 - o Table and/or Flight Software (FSW) Loads
- Contingency Operations and Troubleshooting
 - o Red/yellow limit responses
 - o Document common or ‘expected’ anomalies and their response
 - o Redundancy and redundancy management
 - o Diagnostic modes/activities/data and when appropriate to use

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-3

2. Title:

COMMISSIONING REPORT

3. Reference:

SOW 6.6.6

4. Use:

The purpose of the Commissioning Report is to show that the on-orbit performance will meet the specifications.

5. Preparation Information:

For each activity required to validate on-orbit performance, the following information shall be reported:

- a. Activity information including name, short description, and analysis plan
- b. Test IMT number
- c. Success criteria
- d. Test data used for validation including test date(s) and time
- e. Activity results including activity date and time, data used for validation on the same level as the TVAC reports and where possible show how on-orbit performance compares to ground characterization data
- f. Any test anomalies or idiosyncrasies and their impacts
- g. Health, functionality, and performance

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-4

2. Title:COMMAND AND TELEMETRY (C&T)
HANDBOOK AND C&T DATABASE3. Reference:

SOW 6.4.1.1, 6.5.1, 6.5.2, 6.5.3, 6.6.2.3

4. Use:

Establishes the command and telemetry definitions for the SMBA.

5. Preparation Information:

The Command and Telemetry (C&T) Handbook shall be delivered in Contractor format. The Government prefers two volumes for clarity and ease of use.

The Handbook shall contain the telemetry information. This section of the Handbook shall provide at least, but not be limited to, the following:

- a. Detailed listing of all telemetry assignments and Parameter IDs.
- b. Key parameters necessary for description of the telemetry requirements as a part of the list.
- c. Summary quantifying the number and type of telemetry assignments for each subsystem and the number of spares remaining.
- d. Description of telemetry interfaces, telemetry format, telemetry requirements data, and caution and warning (red and yellow) levels (limits). Note: Modal (state-based) Limits are based on sensor configuration (e.g., subsystem X is not on, therefore subsystem X limits are not checked. Heater Y is on, so limits are ABC. Heater Y is off, so limits are CDE, etc.).
- e. Descriptive information necessary for interpretation of the telemetry requirements.
- f. Listing of telemetry assignments that confirm commands.
- g. Schematic reference for each telemetry assignment.
- h. Data format, to include: detailed bit definitions, command, meta and ancillary data definitions for all telemetry and command data (command and telemetry list), detailed data format and content of the specific Instrument data and ancillary data files as stored on the flight data recorder and specific detailed data formats and outputs of the flight data recorder for Instrument and ancillary data transferred via the Instrument mission data downlink interfaces.

The Handbook shall contain the command information. This section of the Handbook shall provide at least, but not be limited to, the following:

- a. Detailed listing of all serial and digital commands.
- b. Key parameters necessary for description of the commands as part of this listing.
- c. Summary quantifying the number of types of commands used by each subsystem and by each unit and the number of spares remaining
- d. Description of command input, command verification, command rate and filler (no-op) commands.
- e. Description of command requirements data and information necessary for interpretation.
- f. Listing of commands verified by telemetry and the telemetry verifiers.
- g. Schematic reference for each command.

The Handbook shall contain on-board FSW table information and table definitions.

The Instrument Command and Telemetry Database shall be provided in a format agreed to with the spacecraft vendor and operations center.

The telemetry and command database shall contain:

- a) All commands required to operate, monitor, and control the SMBA
- b) All telemetry parameters required to operate, monitor, and control the SMBA
- c) Instrument command and telemetry unique identifiers

Database validation results used to verify the integrity and accuracy of the instrument command and telemetry database shall be delivered with the C&T database.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-5

2. Title:INSTRUMENT LAUNCH AND ON-ORBIT
OPERATIONS AND CONTINGENCY
PROCEDURES3. Reference:

SOW 6.4.1.1, 6.6, 6.6.5.8

4. Use:

The Launch and Early Orbit Procedure section is required to coordinate the launch operation.

The Instrument On-Orbit Procedures section contains the complete set of procedures required for operating the Instrument following completion of commissioning activities. It shall cover all mission phases: activation & checkout, calibration/validation, routine operations, and contingencies.

The Instrument On-Orbit Test and Cal/Val Procedures section of the document contains the set of procedures required to conduct activation, testing, and calibration/validation during the commissioning phase of the mission. These procedures are unique to the activation and commissioning phase and do not duplicate the procedures in the Instrument On-Orbit Operations and Contingency Procedures section. This document shall include the plans, requirements, and procedures for on-orbit sensor calibrations.

5. Preparation Information:

The Instrument Launch and Early Orbit Procedure section of the document shall contain Instrument sequences required to support the Instrument state of health monitoring and operational events during the launch phase of the mission. These procedures are applicable from launch vehicle-Observatory separation until the SMBA Observatory is configured for on-orbit operation prior to the start of the on-orbit system test activities.

The procedure shall include:

- a) Detailed flight timeline
- b) Survival heater configurations (when they need to be turned on), set points, etc.
- c) Script of each communication event including required actions and responses

The Instrument On-Orbit Procedures section of the document shall provide a detailed set of operations procedures for operating the Instrument. It shall provide detailed operating

procedures for each instrument mode, including overview/purpose, prerequisites, applicable mission phases, affected systems/subsystems, flowcharts, commands/mnemonics/parameter values, telemetry mnemonics/verification values, constraints (e.g., time-spacing between commands) and cautions, associated contingencies, duration (if known) and start time (if known). These procedures shall include:

- a) Normal on-orbit command and control operations
- b) Instrument State-of-Health Monitoring and management
- c) Instrument mode transition and mode operations
- d) Contingency and recovery procedures
- e) Calibration procedures
- f) Instrument procedures for satellite reconfigurations

The Instrument On-Orbit Test and Cal/Val Procedures section of the document shall provide a detailed set of operations procedures for activating, testing, and calibration/validation of the instrument. These procedures shall include:

- a) Instrument command and control operations
- b) Instrument State-of-Health Monitoring and management
- c) Instrument mode transition and mode operations
- d) Contingency and recovery procedures
- e) Calibration procedures
- f) Instrument procedures for satellite reconfigurations

Detailed Calibration/Validation Procedures delivered under CV-2 may be referenced from this document instead of duplicated.

All sections, all procedures shall include:

- a) Test/Procedure Purpose
- b) Any Instrument requirement(s) being verified
- c) Procedure Methodology
- d) Support Resources Required
- e) Observatory and Instrument configuration before and after the procedure is executed
- f) Step-by-step commands to be issued and expected Instrument response after each step; including possible contingencies and responses
- g) Cautions and warnings

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-6

2. Title:INSTRUMENT SIMULATOR USERS GUIDE

3. Reference:SOW 4.2.8.2

4. Use:

The Simulator User's Guide describes the operation of simulator for use by the spacecraft contractor

5. Preparation Information:

The Instrument Simulator User's Guide shall contain the information required to use the simulator, including detailed procedures and functionalities. The User's Guide shall show all simulator Graphical User Interfaces (GUIs) and detail the usage of each GUI. It shall give detailed descriptions of major functionality provided by the simulator, and give step-by-step instructions on how to install and use the simulator to achieve these functionalities. The simulator user's guide shall detail the various modes of operation based on access control, and show the difference in screen activations based on a user's input.

The simulator user's guide shall:

- a. List all alerts or notifications produced by the simulator along with their meanings.
- b. Indicate how to start the simulator.
- c. Indicate recovery methods in cases of irrevocable errors or faults.
- d. Indicate data types and command arguments expected within each field of each GUI.
- e. Indicate limitations and differences from the flight instrument.
- f. If necessary, describes any unique configurations or commands associated with each instantiation of the simulator.

DATA ITEM DESCRIPTION

1. CDRL No.:

OO-7

2. Title:COMMAND AND TELEMETRY DATABASE
(CTDB) VALIDATION PLAN3. Reference:

SOW 6.5.1

4. Use:

The CTDB Validation Plan documents the processes used to verify the integrity and accuracy of the instrument command and telemetry database.

5. Preparation Information:

1. The CTDB Validation Plan shall include, but not be limited to, the following information:
 - a. Difference report (e.g., “Was-Is Reports”)
 - b. How difference reports are generated
 - c. Verification of content
 - i. the instrument level database (CDRL OO-4)
 - ii. the system level database (provided by the Government)
 - d. How updates are tracked
 - e. Methods of in-flight or ground-test verification
2. Note: Results from executing the CTDB Validation Plan are provided via CDRL OO-4.

2.5 Project Management Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-1

2. Title:

MONTHLY PROJECT MANAGEMENT
REVIEWS (PMR)

3. Reference:

SOW 1.4, 1.5.1.2, 1.5.1.13, 2.2.1, 4.2.6, 4.2.7

NASA Procedural Requirement (NPR) 7120.5 (Current Revision), NASA Space Flight Program and Project Management Requirements.

4. Use:

To evaluate contract status. These reports will be used to provide an opportunity for face-to-face discussions between the Contractor and the Government regarding project status, plans and issues.

5. Preparation Information:

Scope: The PMR shall include all aspects of the contract effort.

The Monthly PMR shall be presented at a face-to-face meeting with the Government. These meetings shall occur at the Contractor's facility, unless modified by mutual consent.

The Monthly PMR shall include the following:

- A. Program Manager's Assessment and Program Summary
- B. Report of Key Technical Parameters; including mass and power budgets, peak and average power budgets, software metrics, pointing knowledge and control error budgets, their current best estimated, calculated, and/or measured values for all parameters, and trends for each Technical Performance Metric (TPM). The list of reported Key Technical Parameters shall be agreed upon with the Government and may be revised by the Government as the situation dictates. The values to be presented shall be at least to the major component level of the subsystems, e.g., Mechanisms, Power supplies, heaters, cabling, etc. The accuracy of the values and units shall be identified. The status of margins and contingency based on maturity shall be presented. Report Monthly TPM (Current Best Estimate (CBE) and Margin) against the specification for:
 - Mass (Kg)
 - Operational Peak Power (W)
 - Operational Average Power (W)
 - Survival Peak Power (W)

-
- Average Data Rate (Mbps)
 - Peak Data Rate (Mbps)
 - Memory Utilization
 - CPU Utilization
 - NEdT
 - Calibration Accuracy for each channel
 - Non-linearity
 - Gain Stability
 - Center Frequency Stability
 - Bandwidth
 - Out-of-band Rejection
 - Passband Ripple
 - Pointing Accuracy
 - Beamwidth
 - Beam Efficiency
 - Science metrics that are selected to show the satisfaction and stability of the science performance requirements
- C. Technical status for system and subsystem design and development activities, including subcontract technical performance.
- D. Design Status (when design work is being accomplished); covering the design activities for the previous month, including: schedule and technical status of development, manufacturing status, qualification and test status, issues, and concerns
- E. (1) List of Class I in process and approved changes; (2) List of Class II approved Changes available upon request
- F. Summary of Integrated Master Schedule Status, including a brief description of the current status of each subsystem or subassembly along with descriptions of any existing or potential problems areas. The critical path and near critical paths shall be explained along with possible work-arounds being considered to maintain the schedule. A slack metric with at least 1 month of aging (prior month slack) shall be maintained and provided for major subassemblies and activities (the Government and Contractor managers shall determine which items are to be tracked at least quarterly).
- G. Integration & Test Status such as Test Issues, Test Results, Performance Trends
- H. Verification & Validation status and issues (as required)
- I. Mission Assurance status such as summary of MRB actions and status; summary of Failure Review Board (FRB) actions and status; failure investigations and corrective actions
- J. Suppliers Status (as required)
- K. A detailed 12-month “rolling-wave” schedule (3 months of actual, plus 9 months of forecast)
- L. Metrics summarizing all milestones depicting planned versus actual accomplishments and explanations for variances, e.g., graphic display of planned versus actual cumulative milestone status, xx of 1,000 milestones are behind plan and why, etc.
- M. Summary explanation of key Earned Value Management (EVM) parameter performance and trends

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- N. Metrics by Integrated Product Team (IPT) and Total Project, summarizing Baseline, Forecast and Actual Headcounts where a head equals one equivalent work-month of effort. Include explanation of variance from plan.
 - O. Problems (problems that are significant in nature, questionable “significance” should be discussed with and determined by the Instrument Manager or COR) encountered during the reporting period, and anticipated approaches for resolution (including, as appropriate, technical issues including hardware, software, and GSE issues, manpower and staffing, and supplier and subcontractor issues, etc.)
 - P. Status of open issues and problems from prior reporting periods
 - Q. Risk Status for top 10 risks and ability to discuss mitigations plans of each.
 - R. Review of management reserve liens or other unfunded risk mitigations including estimated cost and impact
 - S. Status of action items with focus brought to those actions which are not closing in a timely manner
 - T. PM&P Issues including identification and status of long-lead purchases/acquisitions planned.
 - U. Prohibited Materials Compliance Status (as required)
 - SMBA Product Alert evaluation status
 - Production Readiness Issues (as required)
 - V. Significant plans and activities for the following month, including upcoming milestone events depicting critical items of project status.
 - W. CDRL Status Report that includes the following information for each document delivered in accordance with the CDRL or overdue from previous reporting periods:
 - a. Document Number
 - b. Document Title
 - c. Scheduled Due Date
 - d. Actual Submittal Date
 - e. Current Status
 - f. List of documents planned for delivery during the next reporting period, listed by document number, title, and scheduled submittal date.
 - X. Summary of financial status and Cost Performance per SOW
 - Y. Business issues, including personnel changes
 - Z. Status of Giver/Receiver items
 - AA. Spare parts status

The Contractor shall place PMR material on the Government-access electronic database by the day of the PMR. Presentation material may be in Contractor format.

DATA ITEM DESCRIPTION1. CDRL No.:

PM-2

2. Title:

ENGINEERING PEER REVIEW PLAN

3. Reference:

SOW 2.4, 5.4
GPR 8700.6A, Engineering Peer Reviews

4. Use:

Early in program/product formulation, the Contractor shall identify subsystems, components, software and crosscutting functional elements to be subject to the Engineering Peer Review (EPR) process. The Peer Review Plan is used to identify the methodology and scope of the Contractor's peer review process. The plan shall detail the review process, specifying required personnel, standard agendas, and the procedure for generating, tracking, and closing actions.

5. Preparation Information:

The EPR plan shall be delivered in the Contractor's format.

DATA ITEM DESCRIPTION

1. CDRL No.:

PM-3

2. Title:

INTEGRATED PROGRAM MANAGEMENT
DATA AND ANALYSIS REPORT (IPMDAR)

3. Reference:

SOW 1.2, 1.2.2, 2.2.2.2

Integrated Program Management Data Analysis Report (IPMDAR) Implementation & Tailoring Guide, 24 August 2021

(https://www.acq.osd.mil/asda/ae/ada/ipm/docs/IPMDAR%20Implementation%20and%20Tailoring%20Guide_Stamped.pdf),

Data Item Description for Integrated Program Management Data and Analysis Report (IPMDAR) DI-MGMT-81861C (this document shall be used as guidance in the preparation of the IPMDAR) [ASSIST-QuickSearch Document Details \(dla.mil\)](#)

CPD validator tool [EVM-CR Resources \(osd.mil\)](#)

SPD validator tool [EVM-CR Resources \(osd.mil\)](#)

CPD FFS [20200312 Contract Performance Dataset - FFS.pdf \(osd.mil\)](#)

SPD FFS [20200312 Schedule Performance Dataset - FFS.pdf \(ssitools.com\)](#)

SP-2010-403, [NASA Schedule Management Handbook](#)

[NASA IPMDAR Tailoring Supplement \(NITS\)](#)

4. Use:

To provide information for: (1) integrating cost and schedule performance data with technical performance measures; (2) assessing the magnitude and impact of actual and potential problem areas causing significant cost and schedule variances; and (3) providing valid, timely project status information to higher management.

To ensure an integrated approach to project management, the data provided by this integrated program management data and analysis report (IPMDAR) DRD shall be in consonance with the work breakdown structure (WBS), organizational breakdown structure (OBS), risk management processes, plans and reports (where required), probabilistic risk assessment processes and reports (where required), the cost analysis data requirement (CADRe) (when required), and the monthly/quarterly contractor financial management reports (NASA 533M/Q). The contractor must be able to reconcile between the NASA 533M/Q and the IPMDAR. This reconciliation should be included within the required IPMDAR performance narrative report.

IPMDAR contains data for measuring contractor cost and schedule performance at NASA. The primary purpose is to reflect current performance status and the forecast of future performance.

The IPMDAR consists of the following components:

-
- 1) Contract Performance Dataset (CPD) – Provides performance/execution data from contractor existing management systems.
 - 2) Schedule (Comprised of both Microsoft Project or Native Schedule File and the Schedule Performance Dataset (SPD) – Provides data from the consolidated contractor Integrated Master Schedule (IMS). At a minimum, subcontractor effort must be incorporated into the IMS at an intermediate level with key milestones and be distinguishable from the contractor's and subcontractor's efforts.
 - 3) Performance Narrative Report (Comprised of both the Executive Summary and Detailed Analysis Report) – Provides narrative analysis of data provided in the CPD and the SPD.
-

5. Preparation Information:

The IPMDAR CPD and SPD files shall be validated prior to submittal.

The CPD and SPD formats will be JSON format as defined by the IPMDAR FFSs and the IMS in MS Project file or native schedule format. The IMS data dictionary and performance narrative report will be in Microsoft Word/Excel or native format.

Contract Performance Dataset (CPD), Schedule (Microsoft Project or Native Schedule File and Schedule Performance Dataset (SPD)) and Performance Narrative Report (Executive Summary and Detailed Analysis) Instructions:

5.1 Contract Performance Dataset (CPD), Instructions: Contractor will meet the requirements set forth in [NASA IPMDAR Tailoring Supplement \(NITS\)](#)

The CPD shall be submitted in dollars and hours [choose, delete as appropriate] as part of the original JavaScript Object Notation (JSON) submission. Contractors with IPMDAR flow down may submit dollars or hours [choose, delete as appropriate] as part of the original Prime submission and provide a separate JSON file with dollars or hours [choose, delete as appropriate] file directly to the Government to protect data sensitivities related to rates.

Data Repository-Electronic Submission and File Formats: Data in accordance with the applicable approved CPD FFS and Data Exchange Instructions (DEI) is to include all structures and units to at least the control account/work package level [choose, delete as appropriate].

Subcontractor Integration: Data for Subcontractors with an IPMDAR DRD flow down requirement shall be integrated with the Prime's IPMDAR CDRL delivery.

5.2 Schedule Performance Dataset: Contractor will meet the requirements set forth in [NASA IPMDAR Tailoring Supplement \(NITS\)](#).

GSFC strongly recommends contractors use Microsoft Project as their scheduling tool for integration with GSFC schedules.

Contractors with IPMDAR flow down may submit dollars or hours [choose, delete as appropriate] as part of the original Prime submission and provide a separate JSON file with both dollars and hours file directly to the Government to protect data sensitivities related to rates.

IMS Data Dictionary for Microsoft Project or Native IMS Schedule File’: The first monthly analysis narrative submission for the Schedule shall include the ‘Schedule Assumptions’ and ‘Data Dictionary.’ Any change, post-first submission, to the ‘Schedule Assumptions’ or ‘Data Dictionary’ requires re-submittal of the information with the affected IPMDAR deliverable.

Schedule Risk Assessment (SRA) - Reporting Frequency’: An SRA can be performed, at the request of the Government, particularly when the schedule is a significant issue for the Government due to operational requirements, funding limitations, or when scheduling is considered a major risk element to the acquisition effort.

The IMS contains the contract milestones, accomplishments, criteria, discrete tasks/activities, work packages and planning packages, as applicable, from contract start to contract completion. The IMS is an integrated, logically driven, network-based schedule that is vertically and horizontally traceable. The IMS has traceability to, the project earned value management (EVM) work breakdown structure (WBS), project Organizational Breakdown Structure (OBS), contractor control accounts (CA), and contractor work packages (WP), if applicable to provide transparency for execution. The WBS and other SPD fields in the IMS are consistent with the contract performance dataset (CPD). The IMS contains all calendars that define working and non-working time periods or other information that may impact the schedule. Each event (e.g., milestone, activity, summary, etc.) identified in the IMS has a unique identifier that is consistent period to period.

The Schedule consists of horizontally and vertically integrated discrete tasks/activities, consistent with all authorized work, and relationships necessary for successful contract completion. The Schedule is a single integrated network that also contains significant external interfaces, subcontractor discrete work, Government furnished equipment/information/property and relationship dependencies for the entire contractual effort.

The Schedule shall contain all calendars that define working and nonworking time periods.

The schedule shall reflect accurate remaining durations, start dates, and finish dates for all tasks/activities and milestones in respect to the status date.

5.2.1 If a Statement of Work (SOW) or Integrated Master Plan (IMP) are used for vertical schedule integration, those references shall be provided in both the SPD and native schedule.

5.2.2 Milestones – Provide zero duration schedule events marking the due date for accomplishment of a specified work scope or objective. Milestone may mark the start, an interim step, or the end of one or more activities.

5.2.3 Tasks/Activities - Provide elements of work with duration and logical relationships/dependencies. Task/activity names shall be concise and unique in respect to other

names within the Schedule. The name of each task/activity shall clearly reflect the scope, output (e.g., deliverable), and place within the Schedule architecture so that the content can be understood without the subproject task structure, if applicable.

5.2.4 Duration – Provide the length of time estimated, realized, and/or remaining to accomplish a task/activity. Discrete tasks are preferred to be 42 workdays or less.

5.2.5 Baseline Dates and Information – For all tasks within the performance measurement baseline (PMB), baseline dates in the SPD and Native Schedule files shall be consistent with the start and finish of the cost baseline in the CPD at the control account (or work package) level. The baseline is typically set equal to the current schedule early dates at the time of the baseline establishment when change control is initiated.

5.2.6 Control Account/Work Package Identification – Tasks in the schedule must be tagged with a control account and work package as appropriate. This is accomplished by using the corresponding ControlAccountID and WorkPackageID fields in the Task table of the SPD. The SPD must be consistent with the CA and WP definitions within the CPD. As a minimum, the ControlAccountID values must be consistent with the ID tags as defined in the ControlAccount table and the Task table of the SPD must be consistent with the work package definitions.

5.2.7 Level of Effort (LOE) – Level of effort (LOE) activities may be included or excluded in the network, as appropriate. This determination should be made based on standard procedures. LOE activities shall not impact discrete work or the calculation of the critical and driving paths and never drive paths. When LOE activities are included within the IMS, they are clearly identified as such and defined in the Data Dictionary.

5.2.8 Schedule Percent Complete – The calculated schedule percent complete is a time-based status calculated by the schedule tool without regard to task/activity scope accomplishment. This is not used to status budgeted cost of work performed (BCWP) (i.e., depicts the “time” percent complete based on the forecast completion date, not earned value percent complete based on work accomplished). Earned Value % complete when used, will be in the physical % complete field.

5.1.2.9 Earned Value Technique – The Earned Value Technique (EVT) used for planning and statusing the EV Tool can, as needed, be identified with the Schedule Performance Dataset (SPD). The EVT for each task is identified within the SPD and if LOE tasks are included withing the schedule then they must be identified using the appropriate EVT tag.

5.2.10 Total Float/Slack – Provide the amount of time a task/activity or milestone forecast finish date can slip before delaying contract completion or constraint date.

5.2.11 Free Float/Free Slack – Provide the amount of time a task/activity or milestone can slip before it delays any of its successor tasks/activities or milestones.

5.2.12 Driving Path – The driving path is the longest continuous sequence chain of incomplete discrete tasks/activities/milestones and, if present, parallel chains in the schedule network that

drive the forecast dates of the agreed to discrete task/activity or milestone. A driving path may or may not be on the IMS critical path.

5.2.13 Critical Path – A sequence of discrete work packages and planning packages (or lower level tasks/activities) in the IMS that has the longest total duration with the least amount of total float/slack through an end point.

5.2.14 Subcontractor Tasks – Identify the tasks that are unique to the scope of a major subcontractor, if any. The subcontractor discrete work will be incorporated as tasks within the contractor IMS at a level necessary for a realistic critical path.

5.2.15 Risk Mitigation Tasks – Both the Native Schedule and the SPD delivery shall identify items that came from the Risk/Opportunity Management System and include authorized risk mitigation activities, as applicable.

5.2.16 Schedule Visibility Tasks (SVT) – SVTs are tasks/activities or milestones in the IMS that increase management visibility and functionality of the schedule for non-PMB related items. SVTs shall not be used to represent any scope within the PMB. Resources cannot be assigned to SVTs, nor shall they be used to assess earned value performance.

5.2.17 Lead/Lag – Provide the durations of leads or lags between predecessor and successor tasks. Justification for each lead/lag shall be included in both the Native Schedule and the SPD.

5.2.18 Constraints – Identify the constraints applied to tasks. Justification for each constraint shall be included in the Native Schedule and SPD submissions.

5.2.19 Schedule Margin – A management method for accommodating schedule contingencies. It is a designated risk mitigation placeholder within the schedule and does not have assigned resources.

5.2.20 Data Dictionary for Native Schedule File – Provide a list of all Contractor defined fields, definitions, and code structures used within the Native Schedule File. The Data Dictionary shall be delivered with the initial submission of the Native Schedule File, and resubmitted with subsequent submissions if changes occur to the Data Dictionary, or upon request.

5.2.21 Schedule Risks Assessment (SRA) – The intent of the SRA is to evaluate whether or not the baseline is executable based on the identifiable risks. An SRA is the process which uses statistical techniques to identify technical, programmatic, and schedule risks in a program and quantifies the impact of those risks on the program's schedule.

5.2.22 Custom/User-Defined Fields – Custom/user-defined fields may be required for specific information not otherwise included in the Schedule and included in the data dictionary.

5.2.22.1 Givers/Receivers – Deliverable and receivable products from one organizational element to another organization element must be identified with zero duration milestone tasks in the IMS and finish-to-start dependencies (for example, the deliverable of a subsystem, the

“giver,” to system integration and test [I&T], the “receiver,” must each be a milestone task with zero-day duration linked with a finish-to-start dependency from the giver milestone to the receiver milestone). Giver and receiver milestones must be flagged or coded in the IMS.

5.2.22.2 Government Furnished Item Need Dates – Need dates for Government furnished equipment, material, property, information, facilities, or data must be identified in the IMS with milestone tasks and predecessor and successor logic.

5.2.23 Status Date – Schedule status date must reconcile with the financial and EVM calendar period end dates.

5.3 Performance Narrative Report – Detailed Analysis – Variance Analysis:

Variance Analysis Selection. Variance analysis shall be presented using dollars as the basis.

General: If there are no changes to the reportable element problem analysis, expected impacts, or corrective action status, the contractor may specify “no changes since the last reported analysis” and reference the IPMDAR date when the narrative was reported.

Government Specified Variance Analysis Thresholds. Control account variances that exceed the category thresholds identified below are selected and reported:

Pool of Reportable Variances are determined by the following thresholds:

- 1) Current cost variances exceeding +/- \$100K or +/- 10%
- 2) Current schedule variances exceeding +/- \$100K or +/- 10%
- 3) Cumulative cost variances exceeding +/- \$250K and +/- 10%
- 4) Cumulative schedule variances exceeding +/- \$250K and +/- 10%
- 5) Variances at completion exceeding +/- \$500K and +/- 10%
- 6) Percentages determined using reporting level budgeted cost of work scheduled (BCWS) and BCWP as the base (denominator)

Variance thresholds shall be reviewed periodically to ensure the threshold levels are commensurate with the changing program risks and may be adjusted by contract modification with no change in contract price.

Performance Narrative Report – Detailed Analysis

Detailed Analysis

Required Elements: provide explanations in the Performance Narrative Report Detailed Analysis in accordance with NASA IPMDAR Guide with the additional following definitions:

- Contract summary
- Baseline Changes: Ensure that the following are covered in the Performance Narrative Report Detailed Analysis for baseline changes:
 - 1) Significant Performance Measurement Baseline (PMB) differences (absolute values exceeding +/- 1%) as compared to the previous IPMDAR reported PMB shall be explained in the Performance Narrative Report Detailed Analysis.

- 2) Significant Authorized Baseline Changes shall be explained in the Performance Narrative Report Detailed Analysis. “Significant Authorized Baseline Changes” are defined as the largest baseline changes occurring during the reporting period that in total represent at least 5% (dollar value) of the reporting month’s approved change requests to the PMB.
 - Estimate at completion analysis
 - Undistributed budget analysis
 - Management reserve analysis
 - Cost and schedule variance analysis
 - Supplemental discussions
 - Reconciliation with 533 data
 - Staffing Changes: ‘Major changes’ to staffing are defined as a change of +/- 5% or greater in the reported organizational level staffing as compared to the previous IPMDAR submittal.

The contract IMS shall be reported through delivery of the following information included in the performance narrative report but may be combined at the contractor’s discretion with the approval of NASA’s project scheduling office. All schedule reporting shall be derived from, and traceable to, the contractor’s IMS based on the contractor’s accounting month to include the following:

1. Summary Master Schedule: A one page, top level, Gantt-type schedule chart in contractor format that summarizes the contract effort including period of performance and major milestones, critical path, schedule margin, schedule baseline, and current forecast-to-complete.
2. IMS File/Database: An automated logic network file/database consisting of schedule data for all CWBS elements including subcontractor effort. The entire scope of work must be decomposed into schedule tasks and milestones at a level of detail that supports discrete progress measurement and visibility into the overall design, development, fabrication, integration, assembly, test, and delivery phase of deliverables including flight hardware and software, engineering test and development units, support equipment, etc. Additionally, all schedule tasks must be integrated with dependencies that provide a total end-to-end logic network that includes a system I&T sequence at a level of detail sufficient to identify deliverable need dates. This database must contain dates for all contractual deliverables including end item delivery dates such as flight and ground system hardware and software. Government furnished equipment, material, property, information, facilities, or data need dates must be identified in the IMS. Schedule margin must be clearly defined. The database must contain a coding/flagging scheme necessary to provide sort, select, and summarize capabilities for, but not limited to, CWBS elements, project phase, level-of-effort tasks, and critical path. For contracts that comprise multiple builds, such as more than a single instrument or spacecraft, the IMS must contain the capability to isolate the tasks by unit, serial number, vendor name, or other designation to provide the Government with visibility into baseline schedule, current forecast schedule, and schedule performance.
3. Contractor Schedule Assessment Report: The Contractor Schedule Assessment Report is a narrative report accompanying the monthly IMS deliverable that shall contain the following:

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- a. Schedule Health Check Report: A status on the integrity of the contractor's IMS, based upon an automated schedule analysis tool (e.g., NASA STAT tool, Acumen Fuse, for Project Schedule Analyzer, SSI Tools, etc.) that includes the 14 Point Assessment (DCMA) criteria must be submitted with rationale or corrective actions for health check results that do not meet standard "green" criteria.
 - b. Critical Path Report: This report must be an extract or set of views from the contract IMS and identify the critical path, secondary path, and tertiary path and include all associated tasks and milestones. The report must be submitted in a waterfall format and organized in a manner such that the path with the least amount of slack/float is delineated first, followed by each successive path according to total slack values. This report must contain a narrative explaining changes and impacts to the critical, secondary, and tertiary paths.
 - c. PCM Trend Report: A project control milestone (PCM) cumulative trend chart must be submitted. This report must consist of the baseline PCMs and their corresponding early finish dates, actual PCMs completed to date, and forecast PCMs for the remaining effort and their corresponding late finish dates. The PCM baseline, actual, and forecast dates must trace directly to the IMS. A listing of the PCMs that have not been completed in accordance with the schedule baseline must be included with this report.
 - d. Schedule Margin Trend Report: An accounting of planned and remaining schedule margin displayed in a trend format with explanations for changes during the month reported must be presented.
 - e. Free Slack/Float Report: A report in contractor format must display the free slack/float for each major hardware and software deliverable to system I&T and between one subsystem/element and another. The free slack at the time the schedule baseline was set, the current reporting period's free slack, and the two prior period's free slack/float must be included.
 - f. Recovery and Workaround Plan Narrative: This section of the report must describe any corrective actions, recovery planning, workaround planning, or management intervention being taken or planned to hold or recover from actual and/or potential schedule delays to major contract milestones, PCMs, or givers/receivers.
 4. Schedule Revision Log: The contractor must maintain and deliver a Schedule Revision Log, in contractor format, documenting baseline changes to the IMS. At a minimum, the log should contain the following information: Baseline Change Request (BCR) number, originator, origination date, approval date, date of implementation into the IMS, BCR description and rationale for revision, CWBS affected, and impact of the revision on the overall contractual deliverable.
 5. Weekly I&T Schedule: Beginning one month prior to the start of system I&T activities, a weekly I&T schedule must be submitted in contractor format. The I&T schedule must be delivered on Friday of each week beginning with the start of I&T through completion of the contractor's I&T effort. The I&T schedule must be utilized as the I&T manager's daily working schedule and must contain details of the activities accomplished in the previous week (with the exception of the first delivery) and details of work planned for the next two weeks. The I&T schedule must be organized by shift when appropriate (if more than one standard shift). The weekly I&T schedule deliverable is separate from,

and in addition to, the monthly IMS deliverable. The weekly I&T schedule must be submitted in electronic format (Adobe portable document file is acceptable).

6. Schedule Risk Analysis (SRA): Upon request by the Government, the contractor must provide the results of its probabilistic SRA, including the probability of meeting contractual deliveries and milestones in histogram/S curve formats. The SRA results must also include rank order “tornado diagrams” of both the risks and tasks affecting the completion of contractual milestones and deliverables.

2.6 Review Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-1

2. Title:

PHASE I KICKOFF MEETING DATA
PACKAGE

3. Reference:

SOW 1.5.1.1

4. Use:

To provide a comprehensive overview of the technical approach, design, concept of operations, all Non-Recurring Engineering (NRE) and Safety and Mission Assurance (SMA) processes. Programmatic aspects including program schedule, risk management approach, organizational structure, requirements compliance status with alternatives for any non-compliances, and the specific approach for managing international interfaces. Design establishing a clear baseline of the implementation approach and identifying areas requiring Government attention or decision.

5. Preparation Information:

The Kickoff Meeting Data package shall contain:

- a) An overview and introduction to the organization and personnel (both vendor and Government Teams)
- b) An overview of contractor processes (e.g. Command Media)
- c) The Organizational Breakdown Structure (OBS)
- d) A top-level look at the total program schedule with a more detailed look at the near-term schedule (between kickoff and IBR)
- e) A summary of regularly scheduled meetings and a tentative meeting schedule for the first 6 months (PMRs, table top reviews, peer reviews, program milestone reviews, etc.)
- f) An overview of the instrument design concept
- g) An overview of any technology development work
- h) An overview of current risk database and risk management approach
- i) Overview of configuration management approach
- j) Overview of SMA and quality processes
- k) Overview of vendor facilities (tour if possible)
- l) Proposed Government Mandatory Inspection Points (GMIP) plan for critical items and critical activities

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-2

2. Title:PHASE II KICKOFF MEETING DATA
PACKAGE3. Reference:

SOW 1.5.1.1

4. Use:

To provide a comprehensive overview of the technical approach, design, concept of operations, all Non-Recurring Engineering (NRE) and Safety and Mission Assurance processes. Programmatic aspects including program schedule, risk management approach, organizational structure, requirements compliance status with alternatives for any non-compliances, and the specific approach for managing international interfaces. Design establishing a clear baseline of the implementation approach, and identifying areas requiring Government attention or decision.

5. Preparation Information:

The Kickoff Meeting Data package shall contain:

- a) An overview and introduction to the organization and personnel (both vendor and Government Teams)
- b) The Organizational Breakdown Structure (OBS)
- c) A top-level look at the total program schedule with a more detailed look at the near-term schedule (between kickoff and IBR)
- d) A summary of regularly scheduled meetings and a tentative meeting schedule for the first 6 months (PMRs, table top reviews, peer reviews, program milestone reviews, etc.)
- e) An overview of the instrument design concept
- f) An overview of any technology development work
- g) An overview of current risk database and risk management approach
- h) Overview of configuration management approach
- a) Overview of SMA and quality processes
- i) Overview of vendor facilities (tour if possible)
- j) Proposed Government Mandatory Inspection Points (GMIP) plan for critical items and critical activities

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-3

2. Title:SYSTEM REQUIREMENTS REVIEW
(SRR)/SYSTEM DEFINITION REVIEW (SDR)
DATA PACKAGE3. Reference:SOW 1.5, 1.5.2.1
GSFC-STD-1001A4. Use:

To evaluate the requirements, requirements flow-down, and the operational concepts and to validate the realism of the functional and performance requirements and their congruence with the system configuration selected to conduct the mission. A “delta” review is used to achieve the same, though typically, but not always, on a subset of the design and requirements.

5. Preparation Information:

The System Requirements Review (SRR)/System Definition Review (SDR) shall contain the flow-down of requirements from the Instrument Performance Requirements Document to the Instrument Performance Specification and first-order allocations.

The SRR/SDR Data Package shall contain all relevant instrument information required to satisfy paragraph 4.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The SRR/SDR Data Package shall discuss Contractor system level requirements, rationale, and flow-down plans to lower level requirements.

The SRR/SDR Data Package shall show the allocation and traceability of requirements to major subsystems.

The SRR/SDR Data Package shall include the specification tree illustrating the Contractor’s specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.

The SRR/SDR Data Package shall address any identified Single Point Failures.

The SRR/SDR Data Package shall address redundancy approach to meet mission life and reliability.

The SRR/SDR Data Package shall show how the current concept meets all Government specified requirements including interface requirements.

The SRR/SDR Data Package shall discuss the preliminary operations concept of the instrument.

Where the Government determines a partial Instrument System-level review or a second review is required based on changes in the instrument design baseline, a “delta” review shall be conducted. This “delta” review shall meet the requirements of the appropriate Instrument System-level review (Preliminary Design Review (PDR), Critical Design Review (CDR), etc.) as defined herein and the applicable CDRL. For a “delta” review, the CDRL shall be tailored to cover those items identified for review by the Government.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

Results of Review – As a result of successful completion of the SRR/SDR, the system and its operation are well enough understood to warrant design and acquisition of the end items. Approved specifications for the system, its segments, and preliminary specifications for the design of appropriate functional elements may be released.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-4

2. Title:PRELIMINARY DESIGN REVIEW (PDR) DATA
PACKAGE OR DELTA PDR3. Reference:SOW 1.5, 1.5.2.2, 4.3.1
GSFC-STD-1001A4. Use:

To demonstrate the Flight Equipment and GSE/STE design meet the documented requirements. A “delta” review is used to achieve the same, though typically, but not always, on a subset of the design and requirements.

5. Preparation Information:

The PDR Data Package shall address all relevant instrument information required to satisfy paragraphs 5.2 and 5.3 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The Instrument PDR data package shall contain information to cover the Instrument data processing algorithm design.

The PDR data package shall address the ETU and flight instrument compliance with the Instrument Performance Requirements Document.

The PDR data package shall include responses to action items from previous reviews, including subsystem PDRs/Engineering Peer Reviews.

The PDR data package shall include changes since the last review.

The PDR data package shall address performance requirements and their flow-down to the card or equivalent level.

The PDR data package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The PDR data package shall address error budget determination.

The PDR data package shall address mass, power, data rate, coding and format, commands, EMI/EMC.

The PDR data package shall contain a detailed report of Key Technical Parameters down to a level below the one reported in the PMR.

The PDR data package shall address interface requirements, including the following information:

- a. Preliminary analysis that will allow the Contractor to ensure that the "swept" or deployed volume is verified, accounting for all distortions and misalignments.
- b. Preliminary estimates of gimballed masses and inertias to permit sizing the spacecraft control components to meet pointing and stability requirements.
- c. Preliminary analysis of disturbances of the sum of any periodic disturbance torques, in order to produce the corresponding magnitude spectrum.

The PDR data package shall address mechanical/structural design, analyses, and life tests.

The PDR data package shall address electrical, thermal, optical, radiometric, and calibration design and analyses.

The PDR data package shall address software requirements, design, and development environment.

The PDR data package shall address Ground Support Equipment design and workflow, and describe how each item will be fabricated, tested, and certified when needed.

The PDR data package shall address design verification, test flow and calibration/test plans including any differences between the ETU and flight instrument.

The PDR data package shall address the instrument operations concept.

The PDR data package shall address parts selection, and qualification.

The PDR data package shall address preliminary Failure Modes and Effects Analysis (FMEA); Fault Tree Analysis; and reliability analysis and results.

The PDR data package shall address redundancy and redundancy management.

The PDR data package shall address single point failures.

The PDR Data Package shall address redundancy approach to meet mission life and reliability.

The PDR Data Package shall address the list of long lead items, and of items that may become obsolete prior to completion of all flight instruments, identify those items that must be procured prior to CDR (including a list of those that were ordered prior to PDR and SRR), and provide a plan for procuring these items and all parts.

The PDR data package shall address contamination requirements and control plan.

The PDR data package shall delineate the status of each document required at PDR as to its acceptability for use as is.

The PDR Data Package shall present all instrument risks and address their mitigation.

The PDR Data Package shall provide the status of all sub-contracts and discuss the preliminary design status of critical assemblies and sub-assemblies.

The PDR Data Package shall present a summary of all breadboard and brassboard testing and present the available test results.

The PDR Data Package shall present the development status of sub-assembly engineering units, and available test data.

The PDR Data Package shall present the planned and/or expected level of functionality of the components of the Engineering Development Unit.

The PDR Data Package shall address the producibility of the design solution.

The PDR Data Package shall address mission assurance to be imposed including parts and materials usage as well and workmanship standards imposed.

The PDR Data Package shall address software assurance process.

The PDR Data Package shall include the manufacturing, assembly, integration, and test flow.

The PDR Data Package shall include the specification tree illustrating the Contractor's specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.

Where the Government determines a partial Instrument System-level review or a second review is required based on changes in the instrument design baseline, a "delta" review shall be conducted. This "delta" review shall meet the requirements of the appropriate Instrument System-level review (PDR, CDR, etc..) as defined herein and the applicable CDRL. For a "delta" review, the CDRL shall be tailored to cover those items identified for review by the Government.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-5

2. Title:CRITICAL DESIGN REVIEW (CDR) DATA
PACKAGE OR DELTA CDR3. Reference:SOW 1.5, 1.5.2.3, 4.3.1
GSFC-STD-1001A4. Use:

To present the Instrument Flight Equipment and GSE/STE design and operation, and to demonstrate that all related manufacturing documentation, processes, and fixtures are in place before hardware manufacture of new or improved hardware begins, and to demonstrate that the design meets all performance requirements.

5. Preparation Information:

The CDR Data Package shall contain all relevant instrument information required to satisfy paragraphs 6.2 and 6.3 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The CDR data package shall include responses to action items from previous reviews, including subsystem CDRs/Peer Reviews, and all subsequent subsystem peer reviews held before the CDR.

The CDR data package shall include changes and improvements cited in Section 1.2.7 of the SOW.

The CDR data package shall address ETU and flight instrument compliance with the most recent revision of the Instrument Performance Requirements Document under contract.

The CDR data package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The CDR Data Package shall address the procurement status of long lead items and Electrical, Electronic, and Electromechanical (EEE) parts.

The CDR Data Package shall address manufacturing flow, and the status of the improved manufacturing and assembly drawings, bill of materials, etc.

The CDR Data Package shall address the improved manufacturing procedures.

The CDR Data Package shall address mission assurance product checkpoints and evaluation criteria.

The CDR Data Package shall address standard applicable in-house processes.

The CDR Data Package shall address special/unique to SMBA/fixtures

The CDR Data Package shall address facilities required for manufacturing. The CDR Data Package shall address personnel resources (time phased).

The CDR Data Package shall address the delivery schedules for the ETU, flight hardware and GSE/STE.

The CDR data package shall address detailed analysis from FMEA, fault tree analysis, and reliability analysis.

The CDR Data Package shall address worst case analyses of:

- (a) Electrical circuits
- (b) Scanning drive system
- (c) Lubrication and lubrication loss
- (d) Tolerance and tolerance sensitivity analysis (including thermal and mechanical considerations)

The CDR Data Package shall address stress analyses using NASTRAN with hand verification

The Instrument CDR Data Package shall address thermal analysis of:

- All subsystems (Antenna, Scan Drive, Electronics and Receiver)

The CDR Data Package shall address weight, power, and volume.

The CDR data package shall contain a detailed report of Key Technical Parameters down to a level below the one reported in the PMR.

The CDR Data Package shall address test plans (including all environmental tests) including any differences between test plans for the ETU and flight instruments.

The CDR Data Package shall address manufacturing considerations

The CDR Data Package shall address maintainability considerations, including storage.

The CDR Data Package shall address materials and processes lists

The CDR Data Package shall provide a summary of deviations/waivers

The CDR Data Package shall address contamination control and monitoring considerations including contamination allowances and budgets and methods to ensure necessary contamination mitigation.

The CDR Data Package shall include the manufacturing, assembly, integration, and test flow.

The CDR Data Package shall address spares program.

The CDR Data Package shall address system safety hazards analyses:

- (a) Hazards identification matrix
- (b) Single point failure summaries
- (c) Risk assessment rationale

The CDR Data Package shall delineate the status of each incomplete deliverable document, due at CDR, as to its acceptability for use as is.

The CDR package shall address the operations concept.

The CDR Data Package shall present any additional test results from breadboard and brassboard testing.

The CDR Data Package shall present the test data from sub-assembly engineering development units and the status of ETU development and testing results.

The CDR Data Package shall address the development status of all GSE/STE, including test and calibration procedures, and the software/firmware design and operation and interface aspects as evaluated since the PDR.

The CDR Data Package shall address the status of all program risks and their mitigation.

The CDR Data Package shall address the status of all sub-contract design activities and schedule for delivery of flight hardware, as appropriate, and demonstrate that designs are complete and have been adequately reviewed.

The CDR Data Package shall include the qualification history\status and details for each component for the ETU and flight instruments including data.

The CDR Data Package shall include the specification tree illustrating the Contractor's specifications starting at the top-level Instrument Performance Specification, showing all lower-level specifications to the sub-assembly (box/board) level, and indicating the relationships and interfaces between the documents.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts

package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

Additional Details:

- Implement coax cable and instrument design changes to eliminate blind mates in order to reduce/eliminate risk during cable mate/de-mate operations
- Simplify troubleshooting efforts and reduce risk of hardware damage by eliminating de-integration of instrument components
 1. Implement design changes to allow external measurement of RF ground at the fully assembled instrument level instrument
 2. Provide an assessment on routing the signals for critical subsystem diagnostics to an accessible location at the fully assembled instrument level instrument
- Implement design changes to improve access to W, G, V feed horns as well as improve ease of assembly
- Implement integration fixtures that will be used during receiver shelf integration to eliminate handling errors
- Implement design changes to improve receiver shelf wiring to simplify manufacturing of receiver shelves, implement single point ground, and improve repeatability of design execution
- Automate cable functional tests subsequent to cable builds to reduce handling

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-6

2. Title:PRE-ENVIRONMENTAL REVIEW (PER) DATA
PACKAGE OR DELTA PER3. Reference:SOW 1.5, 1.5.2.4, 5.5
GSFC-STD-1001A4. Use:

To present the description and results of the Instrument Pre-Environmental Test program, and demonstrates readiness for environmental testing.

5. Preparation Information:

The PER Data Package shall contain all relevant instrument information required to satisfy paragraph 9.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The PER Data Package shall include status of action items generated at prior reviews.

The PER Data Package shall include analyses and reports required at the review.

The PER Data Package shall contain the results of any analyses updated or revised since the CDR.

The PER Data Package shall include test and integration program descriptions and results.

The PER Data Package shall include failure report summaries including status of action and rationale for closure.

The PER Data Package shall include as-built documentation summary.

The PER Data Package shall include results of the functional and interface tests.

The PER Data Package shall include descriptions of any malfunctions and corrective actions.

The PER Data Package shall include comparison of measured performance with requirements and discussion of the effect of any variance and waivers.

The PER Data Package shall include mission operation constraints.

The PER Data Package shall include contamination avoidance requirements, recommended practices and approaches for facility contamination control, packaging and protection, and contamination mitigation during test operations.

The PER Data Package shall include safety requirements.

The PER Data Package shall include list of spares for flight equipment and GSE/STE.

The PER Data Package shall include review of flight hardware handling procedures.

The PER Data Package shall include spacecraft interface concerns, problems, and solutions, including:

- a. Analysis and test data that demonstrates that the "swept" or deployed volume is verified, accounting for all distortions and misalignments.
- b. In electronic format, time/magnitude plots of their disturbances, for decomposition, of the sum of the instrument-induced periodic disturbance torques, in order to produce the corresponding magnitude spectrum.

The PER Data Package shall include orbital operations plans and status of documentation and databases.

The PER Data Package shall include end-item data packages (submit a summary of the content prior to review and have package available for inspection at review)

1. As-built configuration list
2. Hardware parts lists
3. Hardware materials and processes lists
4. Test Log Book (including total operating time and cycle records)
5. Open item lists (including reasons for being open)
6. Safety compliance data package
7. Limited life items listings and status
8. Critical parameters trend data
9. Final comprehensive performance test results
10. Failure report summaries including status of action and rationale for closure

The Instrument PER Data Package shall discuss the compatibility of instrument with spacecraft flight support equipment, ground support equipment and operational ground equipment.

The PER Data Package shall address the availability and readiness of facilities and GSE/STE required for environmental testing.

The PER Data Package shall address the status of all program risks and their mitigation plans.

The PER Data Package shall address the readiness of environmental test plans and procedures.

Where the Government determines a partial Instrument System-level review or a second review is required based on changes in the instrument design baseline, a “delta” review shall be conducted. This “delta” review shall meet the requirements of the appropriate Instrument System-level review (PDR, CDR, etc.) as defined herein and the applicable CDRL. For a “delta” review, the CDRL shall be tailored to cover those items identified for review by the Government.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-7

2. Title:

PRE-SHIP REVIEW (PSR) DATA PACKAGE

3. Reference:SOW 1.5, 1.5.2.5, 6.3
GSFC-STD-1001A4. Use:

To evaluate instrument performance during qualification or acceptance testing, and evaluate readiness to ship.

5. Preparation Information:

This data package shall address, as a minimum:

The PSR Data Package shall contain all relevant instrument information required to satisfy paragraph 11.2 (Criteria for Successful Completion) of GSFC-STD-1001A, Criteria for Project Flight Critical Milestone Reviews.

The PSR Data Package shall address responses to action items generated at prior reviews.

The PSR Data Package shall address the solutions to all problems encountered during the environmental test and validation program and the solution rationale.

The PSR Data Package shall address any rework/replacement of hardware, regression testing, and test plan changes.

The PSR Data Package shall address compliance with the test verification matrix.

The PSR Data Package shall address measured test margins versus requirements.

The PSR Data Package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The PSR Data Package shall address qualification/acceptance temperature margins.

The PSR Data Package shall address any data that has been trended to identify compliance with specification, indicating a change or drift to the trend.

The PSR Data Package shall summarize and analyze the monthly trend reports (CDRL SE-2) of trended instrument data, focusing on trends that have changed or drifted anomalously (see SOW paragraph 2.4.2).

The PSR Data Package shall address total failure-free operating time of the item.

The PSR Data Package shall address the number of cycles during testing of parts with finite lifespans.

The PSR Data Package shall address the results of the final audit of any remaining drawing changes.

The PSR Data Package shall address “could-not-duplicate failures” along with assessment of the problem and the residual risk that may be inherent in the item.

The PSR Data Package shall address project assessment of any residual risk.

The PSR Data Package shall provide an update from CDR on shipping containers, monitoring, transportation, and control plans and procedures.

The PSR Data Package shall address ground support equipment status.

The PSR Data Package shall address post shipment plans.

The PSR Data Package shall address spacecraft integration plan.

The PSR shall address the plans for storage of the instrument, if required.

The PSR Data Package shall include documentation verifying that the instrument meets contamination budgets and requirements.

The PSR Data Package shall include alignment data relative to the Instrument Boresight and the mounting surface datum(s) associated with the instrument.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-8

2. Title:

INTEGRATED BASELINE REVIEW (IBR)

3. Reference:

SOW 1.5, 1.5.1.3

4. Use:

To assess and validate the Contractor's Performance Measurement Baseline (PMB) by conducting an IBR.

5. Preparation Information:

The Contractor shall conduct a formal on-site Integrated Baseline Review (IBR). This review will be conducted as early as practicable and conducted in accordance with NASA requirements within 180 days of contract award or formal contract definitization. Additional baseline reviews may be required when substantial changes are made to the baseline (e.g., subsequent detailed planning; including routine rolling wave planning and program re-planning resulting from incorporation of significant contract modifications). The IBR will be tailored to meet the needs of the SMBA project.

The Government will coordinate the timeframe and scope of the IBR with the Contractor. The IBR will ensure that the PMB captures the entire scope of work, is consistent with the contract schedule requirements and has an adequate resource plan. The IBR team shall include Government personnel. Baseline Reviews held during the execution of the Contract will be less comprehensive in nature and specific to the interests of GSFC and the SMBA customer.

The IBR shall cover plans related to the SOW.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-9

2. Title:ENGINEERING PEER REVIEW DATA
PACKAGES3. Reference:

SOW 1.5.1.7

4. Use:

Engineering Peer Reviews (EPR) focus on the design and implementation details at levels that system-level reviews cannot address. They provide a resource for Design Teams to identify potential engineering design and implementation flaws, and increase the probability of success. Applying the EPR process early and throughout the product life cycle affords the maximum advantage in terms of resource efficiency as well as design confirmation and ultimate mission success. Peer review documentation represents knowledge that may prove invaluable later.

5. Preparation Information:

Engineering Peer Review documentation shall be in Contractor format.

The Contractor shall prepare minutes of design reviews that shall consist of a list of meeting attendees, copies of presentation material provided at the review not in the presentation charts package, action items assigned, and agreements made. The reviews will be considered complete when all outstanding action items are closed out to the Government's satisfaction.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-10

2. Title:MANUFACTURING READINESS REVIEW
PACKAGE3. Reference:

SOW 1.4, 1.5, 1.5.1.3, 1.5.1.8, 5.4.1

4. Use:

The purpose of the Manufacturing Readiness Review (MRR) is to verify the engineering baseline by a design, drawing, and engineering requirement review to assure that the release to manufacturing represents the final and correct flight equipment and that the manufacturing planning is complete and reflects the SMBA instrument configuration.

5. Preparation Information:

The content of this review package shall be as follows:

- a) Process flowchart
- b) Schedule
- c) Product description
- d) Configuration Item Identification List (CIIL)
- e) Computer Software Configuration Item (CSCI)
- f) Summary of any changes to configuration baseline, including, parts, materials, and processes.
- g) Summary of any updates to the CDR data submittals, design analysis and SMBA plans and procedures provided under the NASA SMBA Contract.
- h) Detailed manufacturing schedules
- i) Manufacturing flow diagram including inspection, MIPs, and test points
- j) Drawing readiness
- k) Process readiness
 - i. List of drawings
 - ii. Procedures
 - iii. Machine availability and certification
 - iv. Process qualification
 - v. Training and personnel qualification
 - vi. Workmanship standards
- l) Product FMEA (Manufacturing risk mitigation)
- m) Special process list
- n) ESD control plan and cleanliness control
- o) Material availability

- p) Personnel availability
- q) Tooling readiness
- r) Facility readiness
- s) Parts status and qualification status on piece parts
- t) Summary of all nonconformance and test failure reports initiated for the prior flight unit.
The summary shall include the status of all corrective/preventive action.
- u) Status of suppliers
- v) Status of prohibited material screening
- w) A table listing all in-process and end item procedures where CTP, CTB, and/or Consent to Integrate (CTI) decision points require Government approval.

The MRR shall be held as part of a scheduled program review meeting mutually agreed to by the Subcontractor and the Buyer in the baseline schedule; however, where schedule requirements dictate, the MRR may be held at an earlier date. The MRR will be considered approved when all outstanding action items are closed to the Buyer's satisfaction. In the event of disapproval or conditional approval, the required correction(s) shall be submitted to the Buyer for approval. Design review approvals in no way relieve the Subcontractor of responsibility for meeting the subcontract requirements.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-11

2. Title:

DESIGN MODIFICATION REVIEW PACKAGE

3. Reference:SOW 1.5.1.5

4. Use:

For all flight models, to identify changes from the previous model and assess cost/schedule/technical risk. Verify:

- Integration, Test, and Verification approach is adequate
 - Testability items identified that support implementation of the development plan
 - Critical issues and/or activities are identified
 - Risk Items are properly identified with adequate mitigation activities
 - A cohesive plan from the conduct of the review through launch
 - Identify any disconnects, missing items, etc.
 - Identify supplier viability, parts availability, issues with parts obsolescence, facility conflicts, lessons learned.
-

5. Preparation Information:

The Data Package shall include at a minimum:

- Instrument Overview
- Compliance Documents
- By Subsystem/component (hardware and software)
 - Description
 - Documentation Status
 - Supplier – viability, changes in process or product
 - Modifications from previous unit – design, process, qualification approach
 - Test approach
 - Discrepancies
 - Waivers
 - Staffing
 - Risk assessment
- Test Program
 - Plans
 - Procedures
 - Facilities
 - GSE/STE
 - Risk assessment
- Operations products

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-13

2. Title:

TEST READINESS REVIEW (TRR)

3. Reference:SOW 1.5.1.9, 4.2.6
GSFC-STD-1001A4. Use:

A TRR ensures that the test article (hardware/software), test facility, support personnel, and test procedures are ready for testing and data acquisition, reduction, and control. A TRR is held prior to commencement of verification or validation testing. The objectives of the TRR are to:

- Ensure a thorough review of the products supporting the review.
- Ensure the products meet the entrance criteria and success criteria.
- Ensure issues raised during the review are appropriately documented and a plan for resolution is prepared.

A successful TRR signifies that test and safety engineers have certified that preparations are complete, and that the project manager has authorized formal test initiation.

5. Preparation Information:

Prior to holding the TRR the Contractor shall present data to the Government that verifies:

- The objectives of the testing have been clearly defined and documented and all of the test plans, procedures, environment, and the configuration of the test item(s) support those objectives.
- Configuration of the system under test has been defined and agreed to.
- All interfaces have been placed under configuration management or have been defined in accordance with an agreed-to plan, and a version description document has been made available to TRR participants prior to the review.
- All applicable functional, unit-level, subsystem, system, and qualification testing has been conducted successfully.
- All TRR-specific materials such as test plans, test cases, and procedures have been made available to all participants prior to conducting the review.
- All known system discrepancies have been identified and disposed in accordance with an agreed-upon plan.
- All previous design review success criteria and key issues have been satisfied in accordance with an agreed-upon plan.

- All required test resources people (including a designated test director), facilities, test articles, test instrumentation, and other enabling products have been identified and are available to support required tests.
- Roles and responsibilities of all test participants are defined and agreed to.
- Test contingency planning has been accomplished, and all personnel have been trained.

At the TRR and within the TRR data package, the Contract shall show that:

- Adequate test plans are completed and approved for the system under test.
- Adequate identification and coordination of required test resources are complete.
- Previous component, subsystem, and system test results form a satisfactory basis for proceeding into planned tests.
- Risk level is identified and accepted by program/competency leadership as required.
- Plans to capture any lessons learned from the test program are documented.
- The objectives of the testing have been clearly defined and documented, and the review of all the test plans, as well as the procedures, environment, and the configuration of the test item, provide a reasonable expectation that the objectives will be met.
- The test cases have been reviewed and analyzed for expected results, and the results are consistent with the test plans and objectives; test personnel have received appropriate training in test operation and safety procedures.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-14

2. Title:

ETU TEST SUMMARY REVIEW

3. Reference:

SOW 1.5, 1.5.1.4

4. Use:

To present the actual testing conducted compared to the plan, the results and any lessons of that testing, and any subsequent changes, either to the ETU and/or the flight design.

5. Preparation Information:

This data package shall address, as a minimum:

The ETU Test Summary Review Data Package shall address responses to action items generated at prior reviews.

The ETU Test Summary Review Data Package shall address the solutions to all problems encountered during the environmental test and validation program and the solution rationale.

The ETU Test Summary Review Data Package shall address any rework/replacement of hardware, regression testing, and test plan changes.

The ETU Test Summary Review Data Package shall address compliance with the test verification matrix.

The ETU Test Summary Review Data Package shall address measured test margins versus requirements.

The ETU Test Summary Review Data Package shall address instrument performance budgets including Beginning of Life to End of Life performance degradation estimates.

The ETU Test Summary Review Data Package shall address qualification/acceptance temperature margins.

The ETU Test Summary Review Data Package shall address any data that has been trended to identify compliance with specification, indicating a change or drift to the trend.

The ETU Test Summary Review Data Package shall summarize and analyze the monthly trend reports (CDRL SE-2) of trended instrument data, focusing on trends that have changed or drifted anomalously (see SOW paragraph 2.4.2).

The ETU Test Summary Review Data Package shall address total failure-free operating time of the item.

The ETU Test Summary Review Data Package shall address the number of cycles during testing of parts with finite lifespans.

The ETU Test Summary Review Data Package shall address the results of the final audit of any remaining drawing changes.

The ETU Test Summary Review Data Package shall address “could-not-duplicate failures” along with assessment of the problem and the residual risk that may be inherent in the item.

The ETU Test Summary Review Data Package shall address program assessment of any residual risk.

The ETU Test Summary Review Data Package shall provide an update from the ETU Pre-Build Review on shipping containers, monitoring/transportation/control plans.

The ETU Test Summary Review Data Package shall address ground support equipment status.

The ETU Test Summary Review Data Package shall address post-test plans.

The ETU Test Summary Review shall address the plans for storage of the instrument, if required.

The ETU Test Summary Review Data Package shall include documentation verifying that the instrument meets contamination budgets and requirements specified in the Contamination Control Plan.

The ETU Test Summary Review Data Package shall include alignment data relative to the Instrument Boresight and the mounting surface datum(s) associated with the instrument.

DATA ITEM DESCRIPTION

1. CDRL No.:

RE-15

2. Title:INPUTS TO SATELLITE/MISSION LEVEL
REVIEWS

3. Reference:

SOW 1.5.3.1, 1.5.3.2, 1.5.3.3

4. Use:

Inputs to satellite/mission level reviews. To evaluate the requirements, requirements flow-down, and the operational concepts and to validate the realism of the functional and performance requirements and their congruence with the system configuration selected to conduct the mission.

5. Preparation Information:

Inputs to the Satellite/Mission-Level Reviews, including Mission SDR/SRR, Mission PDR, Mission CDR, Mission PER, and Satellite PSR, shall be a subset of the corresponding Instrument level reviews, with appropriate updates.

Inputs to the Satellite Integration Readiness Review (IRR) and Mission System Integration Review (SIR) shall be a subset of the instrument PSR, with appropriate updates.

2.7 Software Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-1

2. Title:

INSTRUMENT SOFTWARE DEVELOPMENT
PLAN

3. Reference:

SOW 1.4, 1.5.1.11, 4.2.1, 4.2.7
NPR 7150.2

4. Use:

Defines Contractor activities required to develop and manage all software, whether developed by the Contractor or by a third party.

5. Related Documents:

NPR 7150.2D

6. Preparation Information:

The Software Management and Development Plan shall describe processes and activities used in the development and testing of the various types of software being acquired, acknowledging the fact that not all software must have the same criticality level or process requirements. Topics to be included in the Software Development Plan are:

- a. Project organizational structure showing authority and responsibility of each organizational unit, including external organizations (e.g., Safety and Mission Assurance, Independent Verification and Validation (IV&V), Technical Authority, NASA Engineering and Safety Center, NASA Safety Center).
- b. The safety criticality and classification of each of the systems and subsystems containing software.
- c. Tailoring compliance matrix for approval by the designated Engineering Technical Authority, if the project has any waivers or deviations to this NPR.
- d. Engineering environment (for development, operation, or maintenance, as applicable), including test environment, library, equipment, facilities, standards, procedures, and tools.

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- e. Work breakdown structure of the life-cycle processes and activities, including the software products, software services, non-deliverable items to be performed, budgets, staffing, acquisition approach, physical resources, software size, and schedules associated with the tasks.
 - f. Management of the quality characteristics of the software products or services.
 - g. Management of safety, security, privacy, and other critical requirements of the software products or services.
 - h. Subcontractor management, including subcontractor selection and involvement between the subcontractor and the acquirer, if any.
 - i. Verification and validation.
 - j. Acquirer involvement.
 - k. User involvement.
 - l. Risk management.
 - m. Security policy.
 - n. Approval required by such means as regulations, required certifications, proprietary, usage, ownership,
 - o. Process for scheduling, tracking, and reporting.
 - p. Training of personnel, including project unique software training needs.
 - q. Software life-cycle model, including description of software integration and hardware/software integration processes, software delivery, and maintenance.
 - r. Configuration management.
 - s. Software documentation tree.
 - t. Software peer review/inspection process of software work products.
 - u. Process for early identification of testing requirements that drive software design decisions (e.g., special system level timing requirements/checkpoint restart).
 - v. Software metrics.
 - w. Content of software documentation to be developed on the project.
 - x. Management, development, and testing approach for handling any COTS, Government-off-the-shelf (GOTS), modified-off-the-shelf (MOTS), reused, or open source software component(s) that are included within a NASA system or subsystem.
 - y. Software Assurance
 - z. Assurance Activities, Tools, and Techniques by discipline:
 - i. Software Quality (process and product)
 - ii. Software Safety
 - iii. Software Reliability
 - iv. Software Verification and Validation
 - v. Independent Verification and Validation (if applicable)
 - aa. Software Assurance Program Metrics
 - bb. Problem Reporting and Corrective Action
 - cc. Assurance records, collection, maintenance, and retention

Note: Verification includes:

- a. Identification of selected software verification methods and criteria across the life cycle (e.g., software peer review/inspections procedures, re-review/inspection criteria, testing procedures).
- b. Identification of selected work products to be verified.

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- c. Description of software verification environments that are to be established for the project (e.g., software testing environment, system testing environment, regression testing environment).
 - d. Identification of where actual software verification records and analysis of the results will be documented (e.g., test records, software peer review/inspection records) and where software verification corrective action will be documented.

Note: Validation includes:

- a. Identification of selected software validation methods and criteria across the life cycle (e.g., prototyping, user groups, simulation, analysis, acceptance testing, operational demonstrations).
- b. Identification of selected work products to be validated.
- c. Description of software validation environments that are to be established for the project (e.g., simulators for operational environment).
- d. Identification of where actual software validation records and analysis of the results will be documented (e.g., user group records, prototyping records, and acceptance testing records) and where software validation corrective action will be documented.

Additionally, the Contractor shall evaluate all flight software using software metrics. The metrics collected, trended, and presented monthly. Metrics shall include at a minimum:

- a. Number of flight software requirements and their change status
- b. Design/Code complexity index at CSU, Computer Software Components (CSC), and Computer Software Configuration Item (CSCI) levels
- c. Source code production rate estimates versus actuals
- d. Number of Software Change Requests/Problem Reports and their status
- e. Resource margins for Utilization of memory, CPU, I/O Bandwidth and Bus traffic
- f. Effort data (staffing profile) estimates versus actuals

Include an alphabetized list of definitions for abbreviations and acronyms used in this document. Include an alphabetized list of definitions for special terms used in the document, i.e., terms used in a sense that differs from or is more specific than the common usage for such terms.

Material that is too detailed or sensitive to be placed in the main body of text may be placed in an appendix or included as a reference. Include the appropriate reference in the main body of the text. Appendices may be bound separately, but are considered to be part of the document and shall be placed under configuration control as such.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-2

2. Title:

SOFTWARE DETAILED DESIGN DOCUMENT

3. Reference:

SOW 1.4,2.4, 4.2.3

4. Use:

The Software Detailed Design (SDD) Document describes the software design and operation for use by software maintenance team. It includes details of the architecture, structure, and organization of a particular Computer Software Configuration Item (CSCI), decomposing the top-level CSCI into Computer Software Components (CSC) and lower levels of units as appropriate. The SDD describes each unit of software in terms of its interfaces (input/output), data architectures, and processing (e.g., logic, algorithms).

5. Related Documents:

NPR 7150.2

6. Preparation Information:

Provide a system level design overview that contains:

- (a) Design Methodology
- (b) Design Overview
- (c) Design Studies
- (d) Design Issues
- (e) Hardware Interface

Provide a system design description that contains, at a minimum:

- (a) Subsystem Description for each subsystem
- (b) Software Description for each software component:
 - a. Description of how the software item satisfies the software requirements, including algorithms, data structures, and functional decomposition.
 - b. Software item I/O description.
 - c. Static/architectural relationship of the software units.
 - d. Concept of execution, including data flow, control flow, and timing.
 - e. Requirements, design, and code traceability.
 - f. CSCI's planned utilization of computer hardware resources.

- (c) Software Interface Control Description for both software-to-software and software-to-hardware interfaces

Describe the system operations design, including:

- (a) Operations Scenarios
- (b) User-System Interface
- (c) Operations Environment and Facilities

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-3

2. Title:FLIGHT SOFTWARE SOURCE CODE AND
VERSION DESCRIPTION DOCUMENT3. Reference:

SOW 4.1.4

4. Use:

To provide the Flight Software Source Code to provide the Government with the capability to build the flight software executable for reference, analysis, and to troubleshoot anomalies.

The Software Version Description Document (VDD) contains a letter describing what is being delivered, the actual executable software, and accompanying documentation. For all firmware media, formatted and numbered drawings are required with assembly numbers assigned to all loaded chips and drawing notes used to identify specific loading and test procedures, markings, etc.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The Flight Software Source Code deliverable shall include:

- a. All items needed to prepare the flight software for the target processor such as: flight software source code, real time operating system, board support package (BSP) software, object oriented models, databases, miscellaneous development tools, utilities, scripts, procedures, and instructions.
- b. The software systems used to develop and maintain the flight software, such as: editors/integrated development environments (IDE), modeling tools, compilers, linkers, loaders, debuggers, and associated configuration management and user interface tools.

The VDD shall include, as a minimum:

- a. Full identification of the system and software (e.g., numbers, titles, abbreviations, version numbers, and release numbers).
- b. Executable software (e.g., batch files, command files, data files, or other software needed to install the software on its target computer).

- c. Software life-cycle data that defines the software product.
- d. Archive and release data.
- e. Instructions for building the executable software, including, for example, the instructions and data for compiling and linking and the procedures used for software recovery, software regeneration, testing, or modification.
- f. Data integrity checks for the executable object code and source code.
- g. Software product files (any files needed to install, build, operate, and maintain the software).
- h. Open change requests and or problem reports, including any workarounds.
- i. Change requests and/or problem reports implemented in the current software version since the last Software Version Description was published.

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-4

2. Title:INSTRUMENT SOFTWARE MAINTENANCE
PLAN3. Reference:

SOW 4.2.10

4. Use:

The Software Maintenance Plan shall contain the information required to sustain the software, including detailed procedures and functionalities. The Plan shall provide insight into the method, approach, responsibility, and processes to be followed for maintenance of software and its associated documentation.

5. Related Documents:

NPR 7150.2

6. Preparation Information:

The Software Maintenance Plan shall include, as a minimum (as per NPR 7150.2D, SWE-105):

- a. Plan information for the following activities:
 - 1) Maintenance process implementation.
 - 2) Problem and modification analysis.
 - 3) Modification implementation.
 - 4) Maintenance review/acceptance.
 - 5) Migration.
 - 6) Software Retirement.
 - 7) Software Assurance.
 - 8) Software Risk Assessment for all changes made during maintenance and operations.
- b. Specific standards, methods, tools, actions, procedures, and responsibilities associated with the maintenance process. In addition, the following elements are included:
 - 1) Development and tracking of required upgrade intervals, including implementation plan.
 - 2) Approach for the scheduling, implementation, and tracking of software upgrades.
 - 3) Equipment and laboratories required for software verification and implementation.
 - 4) Updates to documentation for modified software components.
 - 5) Licensing agreements for software components.

- 6) Plan for and tracking of operational backup software (e.g., backup flight software, backup to the primary operational software).
- 7) Approach for the implementation of modifications to operational software (e.g., testing of software in development laboratory prior to operational use).
- 8) Approach for software delivery process, including distribution to facilities and users of the software products and installation of the software in the target environment (including, but not limited to, spacecraft, simulators, Mission Control Center, and ground operations facilities).
- 9) Approach for providing NASA access to the software version description data (e.g., revision number, licensing agreement).

DATA ITEM DESCRIPTION

1. CDRL No.:

SW-5

2. Title:

INSTRUMENT DATA SETS

3. Reference:

SOW 4.2.11, 6.6.2.3

4. Use:

This data will be used to support ground system development and end-to-end mission rehearsals.

5. Preparation Information:

The sensor test data sets will be determined by the Contractor and **shall** include two sets: one representing normal instrument operation and one with as many different configuration modes and data types as possible.

The Contractor shall prepare and deliver sensor test data sets simulating 72-hour continuous science data.

2.8 Systems Engineering Data Items

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-1

2. Title:

ENGINEERING CHANGE REQUESTS,
DEVIATIONS, AND WAIVERS

3. Reference:

SOW 1.4.1.1, 2.1

4. Use:

To facilitate the orderly processing of engineering change requests to specifications or requirements, specification deviations and waivers, to appropriate level of review, classification, and approval authority for disposition.

5. Preparation Information:

Requests for Waivers and Deviations:

- a. Description of the requirements that apply to the item, subsystem, or system needing the waiver or deviation and provide specific information regarding the nature of the noncompliance.
- b. Description of the effect of the proposed noncompliance on item, subsystem, and system performance at higher levels of assembly.
- c. Description of the cost or schedule impact of non-approval of the waiver or deviation.

Note: Herein, a Class I Engineering Change Proposal is as defined per MIL-HDBK-61B.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-2

2. Title:TREND PLAN/ANALYSIS, REPORTS, AND
OPERATIONS LOG

3. Reference:

SOW 2.5.2

4. Use:

Provides a list and a means of tracking critical engineering and performance parameters for the Instrument and across Instruments. Provides a log of Instrument operating hours.

The trend plan shall include the list of parameters to be monitored for trends leading toward degradation of performance or reliability of the Instrument and out of family with other Instruments. Starting at component acceptance testing and continuing during the system integration and test phases through the on-orbit commissioning phase, trend data shall be collected and analyzed in trend reports. The trend list shall be revised with the on-orbit operational trend parameters prior to launch. An operations log shall be maintained of the accumulated operating time of the Instrument.

5. Preparation Information:

The trend plan, trend reports, and operations log shall be delivered in Contractor's format.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-3

2. Title:

ENGINEERING DRAWINGS

3. Reference:

SOW 1.4, 1.4.1.2

4. Use:

To provide layouts and engineering drawings to serve as the basis for technical discussions, evaluations, manufacturing, fabrication, assembly, test, operations, and maintenance. To satisfy some of the Observatory drawings requirements of the launch services provider.

5. Preparation Information:

The Contractor shall maintain an indentured drawing list of all drawings and make that list available on request.

The Contractor shall make available on request, engineering drawings used to procure, manufacture, assemble, integrate, test and control interfaces for flight and GSE. These include all reference type drawings such as layouts, schematics, diagrams, mechanical drawings, electrical schematics, logic diagrams, and block diagrams.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-4

2. Title:SYSTEM PERFORMANCE VERIFICATION
PLAN AND REQUIREMENT VERIFICATION
MATRIX3. Reference:

SOW 2.2.2.1, 5.7.1

4. Use:

Provides the overall approach for accomplishing the verification program. Defines the specific tests, analyses, calibrations, alignments, hardware models, etc., that will demonstrate that the flight hardware complies with the mission requirements.

5. Preparation Information:

The SPVP shall:

- a. Provide an overview of the entire Instrument verification program.
- b. Flow performance requirements to all levels of assembly and describe the verification method for these requirements/specifications.

The SPVP describes the approach (test, analysis, etc.) that will be utilized to verify each requirement in the SMBA Instrument Performance Requirements Document (PRD) and the Spacecraft to SMBA ICD. Verification will be to the BOL performance levels in the SMBA Instrument Performance Specification. If verification relies on measurements, tests, or analyses at lower (or other) levels of assembly, this dependence shall be described.

The SPVP shall document plans to verify by test or inspection the proper polarity, orientation, and position of all components (sensors, switches, and mechanisms).

This Plan shall include level of assembly, configuration of item, objectives, test phases, appropriate functional operations, and requirements for procedures and reports. Each analysis activity shall include objectives, a description of the mathematical model, assumptions on which the model will be based, required output, criteria for assessing the acceptability of the results, interaction with related test activity, and requirements for reports.

On-orbit verification tests shall be cross-referenced to the On-Orbit Commissioning Plan (CDRL OO-1).

The Final SPVP due 15 days prior to PER shall incorporate all review comments.

Requirement Verification Matrix (RVM)

The RVM shall summarize the flow-down of system specification, Interface, Mission Assurance, and calibration/validation requirements verification. The RVM shall stipulate how each requirement (referenced by requirement ID) will be verified, the expected date for closure, the owner of the requirement, and summarizes current status of compliance/non-compliance with requirements. The SPVM shall list a summary description of each requirement, and a summary of the measured/analyzed/demonstrated performance of the system against each requirement. It shall show each SMBA Instrument Performance Requirements Document (as flowed to the SMBA Instrument Performance Specification), Spacecraft to SMBA ICD, and Calibration/Validation Plan requirement reference source (to the specific paragraph or line item), the method of compliance, applicable procedure references, report reference numbers, etc., for each requirement set from the SMBA Instrument Performance Requirements Document. It shall show the flow-down of requirements verification through the sub-system (box/board) level.

The RVM shall include the level of assembly at which the requirement is verified and the environments in which the requirements will be verified. Requirements which cannot be verified at the instrument level (i.e., must be verified at observatory level) shall be clearly identified.

The RVM shall trace requirements backwards to the next level above, i.e., a Level 4 requirement shall be traced back to its Level 3 parent, etc.

To allow transfer/tracking into DOORS, the RVM shall be prepared and delivered in DOORS compatible (electronic) format (Excel is acceptable), to include, at a minimum, the following content:

- Source of Requirement (e.g., PRD, Inst to SC Interface Requirements Document (IRD), etc.)
- Requirement Reference ID (e.g., PRD-1446, SCIRD-0001, etc.)
- Requirement Text
- Verification Method (e.g., test, analysis, etc.)
- Test Plan
- Test Procedure (number and title)
- Verification Lead
- IMS ID
- Expected Date of Closure (Milestone)
- Verification Status
- Artifact report (e.g., Test Report, Analysis Report, etc.) (number and title)
 - NOTE: Any artifacts not formally delivered shall be made available for review on request

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-5

2. Title:INSTRUMENT PERFORMANCE
SPECIFICATION

3. Reference:

SOW 2.2, 2.4, 2.5.1, 4.2.2, 4.2.3

4. Use:

To document the instrument system and subsystem specification derived from the Instrument PRD and the other contract requirements documents to ensure that the Instrument design requirements flow correctly from the Instrument Performance Requirements Document and the other contract requirements documents to lower level requirements. The instrument performance specification shall indicate the flow down and allocation of all SMBA technical (functional and performance) requirements to lower levels (including subsystems, software, etc.).

5. Preparation Information:

The SMBA Instrument Performance Specification shall be delivered in the Contractor specified format.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6

2. Title:

ANALYSES AND MODELING

3. Reference:

SOW 2.2, 2.5, 2.5.1

4. Use:

To provide reports of specific analyses for review.

5. Preparation Information:

All models shall be exchanged in formats that are compatible with those used by the spacecraft Contractor.

Sufficient model documentation shall be provided to describe how the significant thermal and structural characteristics are modeled.

Nodal number ranges and element number ranges should be defined for the individual components that make up the model.

The model will be utilized by the Spacecraft Prime Contractor for system-level modeling.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6a

2. Title:STRUCTURAL AND MECHANICAL
ANALYSIS AND MODELS3. Reference:

SOW 2.4, 2.5.1.1

4. Use:

To document the analyses and data which demonstrates that all structural and mechanical subsystem performance requirements are satisfied. These models are to be used to analyze the structural and dynamic characteristics of each instrument. The models will be used to determine the instrument structural adequacy to withstand transportation, launch, and on-orbit loads. Also, they will be used to predict instrument structural resonant frequencies. The satellite/launch vehicle loads analysis will be used in the development and definition of the instrument/spacecraft interface loads. The model will be utilized by the Spacecraft Prime Contractor for system-level modeling.

5. Preparation Information:

This analysis and report shall contain the following in compliance with applicable requirements as specified in the ICD:

The Contractor shall provide documented verification that all structural and dynamic requirements of the equipment are met. The Contractor shall utilize analytical models and perform the simulations necessary to support and justify the design approach and solution.

The structural analysis report(s) shall include, but shall not be limited to, the stress analysis of critical structural elements of the chassis, dynamic response of the printed wiring boards and chassis walls containing subassemblies, and deflection analysis. The analysis report should contain, at a minimum: all assumptions and design criteria used in the analysis, descriptions of the analytical models, and results, conclusions, and recommendations. All detail calculations necessary to substantiate results; computer detailed calculations other than standard outputs are exempt. Decompression/venting analysis of the equipment shall also be included.

The Contractor shall verify the accuracy of the models with dynamic test data. The Contractor shall update the models to agree with the structural test results. A structural analysis using

maximum equivalent loads shall be conducted on the instrument. The Contractor shall perform a structural analysis on each separately-mounted instrument component.

NASTRAN Finite Element Models (FEM) of the Instrument shall be delivered by the Contractor. A test-verified detailed finite element model shall be provided if the instrument exceeds 23 kilograms and the fundamental resonant mode frequency is less than 50 Hz, as shown by analysis.

The Contractor shall provide analysis and test data that describes the "swept" or deployed volume which accounts for all distortions and misalignments.

A cross orthogonality check between test and analytical mode shapes should have diagonal terms above 0.9 and off diagonal terms below 0.1 for all significant modes up to 50 Hz.

Instrument alignment and stiffness: This data item shall document the analyses, tests and test data required to verify all instrument alignment and stiffness requirements. The document shall also address instrument mechanical design verification with respect to dynamic interactions with the launch vehicle.

Instrument clearances: This data item shall document all instrument critical clearances and loss of clearances, with the associated analyses. This shall include dynamic loss of clearance between the observatory and the launch vehicle, and between various observatory elements. Clearance loss shall be determined for critical clearances throughout all mission phases, including launch, on-orbit deployments, and on-orbit operations. Components contributing to each clearance loss, such as manufacturing and assembly tolerances, clearance loss due to observatory dynamics and thermal gradients, as well as observatory insulation, harnesses, and grounding provisions, shall be quantified. Analytical and measured data used for clearance verification shall be provided.

Mechanical performance: This data item shall fully document the tests and analyses performed to verify the performance of the instrument mechanisms and deployment devices. Analyses and test data used to verify the instrument structural performance and workmanship shall also be provided. This document shall also address any methods utilized during fabrication and assembly which verify workmanship of the observatory structure and mechanical components.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6b

2. Title:

THERMAL ANALYSIS MODELS

3. Reference:

SOW 2.4, 2.5.1.2,

4. Use:

To document the detailed and reduced thermal analyses used in the design of the instrument. Several analysis reports and the instrument thermal models themselves are required to be delivered as noted in the table of CDRL deliverable models and reports.

The instrument reduced thermal model will be integrated with the satellite thermal model. The satellite thermal model will be utilized by the Spacecraft Prime Contractor to characterize satellite thermal performance. The satellite thermal model will be used to calculate radiation exchange factors, environmental heating rates (solar, albedo, earth infrared) during various phases of the mission including launch, ascent, on-orbit operations, all survival modes, and including instrument calibration maneuvers. The satellite thermal model will be used to characterize the thermal response of the satellite for all spacecraft states and instrument modes over the entire length of the mission. The satellite thermal model will also be used to simulate the thermal environments encountered during the satellite environmental test campaign.

5. Preparation Information:

The analyses shall contain, as a minimum, the following: all pertinent assumptions used for analysis, description of the thermal model with associated boundary conditions, all detailed calculations needed to substantiate the results (if used), the temperature and mechanical effects on the unit in thermal vacuum (if required) versus ambient pressure, and results, conclusions, and recommendations. References/measurements shall be cited to support assumptions about interface conductances and optical and thermophysical properties.

The detailed thermal analysis shall demonstrate that all internal parts meet established thermal design criteria, based upon program de-rating or electrical performance requirements. The associated reports shall describe in detail the various analysis cases (e.g., launch, cold survival, worst case hot and cold operational, nominal science, etc.). This description shall include:

- a. List of all parameters and assumptions
- b. Relationships between various model, or sub-models, and how boundary conditions are obtained
- c. Detailed calculations and descriptions of computer models and analyses

- d. Results of the final analyses including predicted thermal performance for worst case extremes and design margins

Instrument Reduced Thermal Model Requirements:

The Instrument Contractor shall provide a reduced thermal model using ThermalDesktop (geometry) and SINDA/FLUINT (math model).

The thermal model shall include an adequate level of detail to predict, under worst case hot and cold environments, all critical temperatures, including those that drive operational and survival temperature limits and heater power. Worst case conditions shall include variations in season, orbit selection, orbital time, and environmental flux parameters (seasonal and spatial) and a rational combination of the effects of design tolerances, fabrication uncertainties, material differences and degradation due to aging. The instrument thermal model will also be used to support thermal distortion studies, pointing budget assessments and/or power budget activities as required. The geometry used to represent instrument radiators, apertures, optical elements, mounting feet, and any other components identified as 'critical' in the thermal model shall be congruent with the geometry found in the mechanical/CAD model. Instrument thermal models may deviate from their mechanical counterparts to account for soft goods, such as multi-layer insulation blankets, or to intentionally exclude unnecessary features.

For each flight temperature sensor, thermostat, heater, or other active thermal control element (e.g., thermoelectric cooler), a thermal model node(s) shall be used to represent the corresponding hardware location.

For purposes of this DID, instrument 'critical components' are defined as comprising the hardware elements cited in the mechanical/CAD model and the thermal model node(s) shall be used to represent the corresponding hardware location.

Instrument thermal models shall employ closed volumes and/or ThermalDesktop Radiation Analysis Groups to separate radiative exchanges involving interior instrument surfaces from radiative exchanges involving exterior instrument and/or spacecraft surfaces. It is recommended that radiative exchanges between the instrument base plate and the spacecraft, unless intentionally retained, be zeroed by specifying the base plate surface opposite the spacecraft interface as 'Active None' (ThermalDesktop feature).

Models shall use conservative property values for conduction, absorption, emission and MLI effective emittance, and consider contact resistance. Over time, some of these conservative values are expected to give way to direct measurements and model correlation findings. Analyses should consider beginning of life and end of life properties for external coatings.

All instrument thermal models shall be capable of steady-state and transient analyses. To satisfy the latter portion of this requirement, thermal models must accurately account for the measured/anticipated instrument mass.

Instrument thermal models shall use the following units: centimeters, kilograms, Joules, seconds, and degrees Celsius.

Thermal models shall be updated as necessary to reflect significant design changes during the design phase of the instrument, and analyses used to support such design changes shall be revisited using the test-correlated detailed thermal model on completion of instrument thermal balance testing.

Instrument reduced thermal models, constructed using ThermalDesktop and SINDA/FLUINT, shall be provided by the Instrument Contractor for use by the Spacecraft Prime Contractor in defining each instrument/spacecraft thermal interface. The reduced thermal model will be integrated into the observatory thermal model by the Spacecraft Prime Contractor for observatory thermal analyses.

All models shall be fully documented to permit ease of use by other contractors as deemed appropriate by the Program.

The reduced thermal model shall be as small as is reasonably possible. The total number of surfaces is dependent on geometric complexity and the maximum acceptable surface count shall be coordinated with the Spacecraft Prime Contractor and the Program.

All reduced thermal model SINDA/FLUINT entities shall reference SINDA/FLUINT Submodel names agreed to by the Instrument Contractor and the Spacecraft Prime Contractor. Once the instrument detailed thermal model correlation is complete, the reduced thermal model shall be updated to reflect any corrections made to the detailed thermal model. In creating the observatory thermal model, the Spacecraft Prime Contractor will place the instrument reduced thermal models in accordance with the appropriate mechanical interface document (MICD). To support this effort, the Instrument Contractor shall specify the precise relationship between the reduced thermal model geometry and the instrument-to-spacecraft MICD.

When the post-correlation reduced and detailed models are subjected to the same environmental heat loads and internal dissipations, the reduced thermal model shall reproduce detailed thermal model temperature predictions for all critical components to within 3C. When the reduced and detailed models are subjected to the same environmental heat loads and internal dissipations, the reduced thermal model shall reproduce detailed thermal model instrument- spacecraft heat flows to within 5%.

Instrument Thermal Model Documentation:

The following information and data shall be supplied in fulfillment of the requirement of a deliverable instrument thermal model:

- a. Description of the model, any special modeling features used, and the rationale for the modeling methodology.
- b. Illustrations/diagrams of the energy balances, also known as 'heat maps,' simulated by the thermal model.

-
- c. A summary of the organizational scheme used to group thermal model entities to represent instrument components and/or modules (e.g., an optical module might be identified by using a distinct SINDA/FLUINT submodel, a specific set of node/element IDs, and/or through the use of ThermalDesktop Groups and Layers).
 - d. ThermalDesktop verification plots showing the locations of thermal blankets, radiators, apertures, and thermal control hardware.
 - e. A description of any thermal control hardware (e.g., radiators, thermal blankets, heaters, temperature sensors), apertures, optical elements, conductive interfaces (e.g., mounting feet), and any other components identified as 'critical' hardware and the key parameters associated with this hardware. A 400 in2 aluminum radiator, for instance, might be covered with a 5 mil thick silver Teflon film and have a mass of 1kg. Or, a titanium instrument flexure might have a 0.3 in2 minimum section, 0.25 W/K conductance and be secured to the spacecraft mounting plate by 2, M5 bolts. The thermal model correlation effort shall focus on adjusting the detailed thermal model to matching test data for these items.
 - f. Any instrument component represented only by a node(s) (i.e., there is no corresponding surface/solid geometry in the thermal model) shall be identified (e.g., node number 1967), described (e.g., fold mirror) and characterized (i.e., predominate material, mass, dissipation) in the associated thermal model report(s). Such entities might be found in SINDA/FLUINT INCLUDE or INSERT files called by the ThermalDesktop Case Manager.
 - g. A table describing the estimated, or measured, heat dissipated by the instrument for all environmental test or flight modes and physical configuration. If the dissipation for an instrument with a staring telescope is different than it is with scanning telescope, for instance, thermal dissipation estimates are required for both scenarios.
 - h. A table listing beginning-of-life and end-of-life assumed and/or measured optical properties (i.e., solar absorptivity and IR emissivity).
 - i. A table listing thermophysical properties (e.g., density, specific heat, conductivity) of the materials used to create instrument components.
 - j. References shall be provided to support all assumptions regarding optical properties, thermophysical properties, multilayer insulation performance and joint conductances.
 - k. Instrument Thermal Analysis report shall describe in detail the various analysis cases (e.g., launch, transfer orbit, on-orbit, on orbit hot and cold cases, etc.). This description shall include:
 - a) List of all parameters and assumptions used
 - b) Interrelationships between various models and how boundary conditions are obtained
 - c) Detailed calculations and descriptions of computer models and analyses
 - d) Results of final analyses including predicted thermal performance over worst case extremes and design margins
 - e) General outline of the report should include:
 - a. Purpose
 - b. Related documentation
 - c. Requirements Verification
 - i. Which requirements are verified through this report
 - d. Thermal Model Description

- i. Component Models
 - ii. Thermal Optical Properties
 - iii. Thermophysical Properties
 - iv. MLI Effective Emissivity
 - v. Orbits
 - vi. Component Power Dissipations
 - vii. Internal and External Thermal Isolations
 - viii. Spacecraft interfaces
 - ix. Heater Voltages
- e. Thermal Analysis Results
 - i. Operational Case Temperature Predictions
 - ii. Survival, Decontam, and Safehold Cases Temperature Predictions
 - iii. Temperature Margins
 - iv. Thermal Interface Rate of Change Predictions
 - v. Thermal Gradients (if required)
 - vi. Heater Power Predictions
 - vii. Heat Flow Predictions (Heat Map)
 - viii. Launch Analysis
 - ix. Any other special analysis not covered in sections above
- l. Sample runs characterizing worst case hot and cold operating environments and worst case hot and cold survival environments shall be provided.
- m. For the cases anticipated by the worst case hot and cold operating environments and worst case hot and cold survival environments, tables comparing detailed and reduced thermal model temperature predictions and heater duty cycles shall be provided.
- n. For each node used to represent a flight temperature sensor a temperature correction shall be provided.

DATA ITEM DESCRIPTION1. CDRL No.:

SE-6c

2. Title:

CONTAMINATION ANALYSIS REPORT

3. Reference:

SOW 2.3,

4. Use:

To document contamination analysis for Government review.

5. Preparation Information:

The primary focus of the instrument internal mass transport analysis is an understanding of the potential contamination sources within the instrument, as well as a list of surfaces sensitive to contamination depositions. The results should include a list of the deposition onto sensitive surfaces (usually cold surfaces) for the duration of mission life as well as the temperature history of these elements.

A list of the surfaces of concern as contamination sources should also be presented as a result. This allows the engineering team to evaluate the design to see if there are design changes which could limit these effects (e.g., lower source temperatures, baffles, sealed areas). The results can also help determine the necessary levels of bakeout required to insure proper performance of the instrument. In addition, a thorough understanding of the outgassing paths within the instrument can help understand contamination aspects of potential flight problems that may be contamination related.

Instrument level TV test outgassing results will further confirm or refine these models.

Information required for NEON to perform analysis:

If the instrument vendor is unable to perform the necessary mass transport analysis, the NEON mission may be able to provide support. If an agreement can be reached to allow this assistance, the following information is required for analysis to be performed. If this information is not provided, NEON will not be able to adequately perform the analysis.

A STEP and/or ThermalDesktop file for geometry is required. If these are not available, up to date drawings may be able to be used to develop this model however this will delay the performance of the analysis. In addition, the material used to build the hardware and the materials properties are also required. Materials selection and location is a key component in determining where contaminants will move.

A file of the temperature predicts for a typical hot and a cold case for each node including bus, instruments, and other exposed hardware. Each component must specify radiator location(s) and vent size/location for all hardware. The maximum amount of contaminants allowed onto sensitive surfaces and apertures for the duration of mission life in order to satisfy throughput requirements for optics is also required.

For internal hardware analysis, a reasonably detailed geometry file for internal layout of the component (e.g., how many cavities inside the instrument, baffles, venting path, vent location to outside, identify contamination sensitive surfaces), typical operating temperatures for all of the nodes, and outgassing rates of each component are all required.

Requirements for chamber bakeout performance analysis:

If the analysis is to determine the bakeout performance criteria for a hardware outgassing certification, even more information is required. In addition to all of the above data, the layout and operating parameters of vacuum chamber are also required. A detailed layout of the arrangement of the hardware within the chamber is essential. Temperatures of all chamber components, and locations of all vents and cold traps are also required. The location and operating temperature of the Thermal Quartz Crystal Microbalances (TQCM) used to monitor the bakeout/certification must also be specified. In addition, the chamber pumping efficiency must also be determined by the vendor.

The chamber pumping efficiency is a function correlating the amount of contaminants which are pulled out of the chamber by the pump and/or cold traps versus the amount of contaminants which will be deposited on the TQCM head. This function is usually calculated by performing a chamber background bake under two different conditions. For the first portion of the test, the background should be taken with no cold traps operating. Once at least 8 hours have passed under these conditions, a cold trap of well-known temperature and area should be run for at least 8 hours. This data is required to then calculate the pumping efficiency for the chamber analysis.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6d

2. Title:

EMI/EMC ANALYSIS REPORT

3. Reference:

SOW 2.5.1.4
PRD 5.7.5, 6.2, 7.3
MIL-STD-461G

4. Use:

To document the EMI/EMC analyses used in the design of the instrument.

5. Preparation Information:

To be provided in Contractor preferred electronic format.

The Contractor shall document the EMI/EMC analyses used in the design of the instrument as defined in the instrument PRD to verify the hardware requirements.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6e

2. Title:

HYPERSPECTRAL CAPABILITY ANALYSES

3. Reference:

SOW 2.5.1.8, 5.3

4. Use:

To document the characteristics and performance of the SMBA instrument's hyperspectral capability.

5. Preparation Information:

The delivery shall include:

1. A description of the hyperspectral frequency configuration:
 - a. Identification of which bands and frequency ranges have hyperspectral capability
 - b. The number of channels for each hyperspectral band
 - c. The start and stop frequencies, the center frequency and bandwidth of each hyperspectral channel
 - d. Performance trades and studies used to select the hyperspectral configuration, including rationale on the selected configuration
2. The performance characteristics of the hyperspectral channels:
 - a. The Spectral Response Function (SRF) of each channel, including a description of which passbands are combined to form the channel (a reference to CV-10 SRFs is acceptable)
 - b. The raw data used to derive the SRF along with the frequencies associated
 - c. The description of the SRF measurement test that provides details about the test procedure (set up and instrumentation), the environmental conditions, the data generated, the data analysis and the results obtained
 - d. A description of the algorithms that combine the channels including rationale; limitations and specifications; mathematical description; a performance analysis; and assumptions such as the data sets used for simulating the environmental conditions
 - e. The start and stop frequencies, the center frequency stability, calibration accuracy, and NEdT of each channel
3. Observatory- and Mission-level considerations:
 - a. Flexibility in the frequency plan, including capabilities to update the frequencies pre- and post-launch
 - b. Power consumption impacts

- c. Thermal impacts
 - d. Data rate impacts on downlink
 - e. Data storage impacts
- 4. Hyperspectral subject matter expert inputs:
 - a. Description of the signal flow and digitization, including Fast Fourier Transforms, binning, windowing, and similar functions as applicable
 - b. A description of the key contributing hardware elements and their functions
 - c. A description of the software
 - d. Digitization errors, including how closely a digitized signal matches the analog signal

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-6f

2. Title:RADIO FREQUENCY INTERFERENCE
DETECTION CAPABILITY ANALYSIS3. Reference:SOW 2.5.1.7, 5.2
MIL-STD-461F4. Use:

To document the capability for the SMBA instrument to detect radio frequency interference from anthropogenic RF sources in the SMBA detector frequency range. Because the frequency range of the instrument detector is near the range used for RF communications it is important for the instrument to be able to identify and flag RF interference that may contaminate science data measurements. This analysis documents the methods and capabilities the SMBA instrument will use to identify RF interference.

5. Preparation Information:

1. The delivery shall consist of the description of the Radio Frequency detection capability of the SMBA instrument covering both the hardware and software aspects for each of the bands identified in PRD Table 4.1-2, Hyperspectral and RFI Detection Capabilities by Band. It shall include:

- The number of channels for each band
- The bandwidth of each channel
- The Radio Frequency Interference (RFI) detection algorithms chosen for each band
- The implementation of the RFI detection algorithms

For each of the selected RFI detection algorithm, the delivery shall consist of:

- An overview of each algorithm including the objective or purpose of the algorithm
- The rationale for each algorithm and identification of any specifications and limitations
- A description of each algorithm's inputs and outputs including any external ancillary data (i.e., thresholds) used by the algorithm
- A mathematical description of each algorithm
- An implementation of the algorithm in source code and identification of the Operating System that the source code was built on

- A performance analysis of the algorithm that provides metrics on the algorithm including its detection sensitivity, its probability of detection and its false alarm rate as function of RFI characteristics
- A description of the methods used to validate each algorithm and description of the validation results

At a minimum, RFI test cases shall include RFI algorithm capability assessment for each of the following categories of RFI:

- a) Continuous Wave (CW) with and without modulation
- b) Pulse wave with various duty cycles (5% ~ 60% duty cycle)
- c) Narrow band and wide band interference with various bandwidth (5-400MHz bandwidth interferes based on 5G OFDM spectrum and similar)
- d) Multiple sources with combinations of a, b, and/or c

All RFI test cases should be simulated and tested with a range of Interference to Noise Ratio (INR) in dB in addition to representative of diverse natural backgrounds. For hard decision RFI detection algorithm (yes/no), Contractor shall provide the Receiver Operating Characteristic (ROC) curve where each point represents different INR. For soft decision (selectable threshold), Contractor shall provide for each INR case a plot of the ROC curve where each point represents different threshold.

A test report that describes the procedure used to test the SMBA instrument's RFI detection capability for each of the bands identified in PRD Table 4.1-2, Hyperspectral and RFI Detection Capabilities by Band, shall be delivered. This test report shall include a description of:

- The RFI test set up and instrumentation
- The RFI characteristics tested
- The RFI detection algorithms tested
- The data sets generated by the test of the RFI detection algorithms
- Test Objectives
- Test Methods
- Assumptions
- Test Equipment Identification
- Environmental and/or other conditions to be maintained
- Test durations
- Test configurations
- Sequence of events
- Step-by-step instructions
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Analysis techniques
- Expected results
- Pass/Fail Criteria
- Test Termination Procedure

A test report that describes the procedure used for the RFI test cases reproducing what is currently observed from both terrestrial and space-based RFI sources shall be delivered. This test report shall include a description of:

- The RFI test set up and instrumentation
- The RFI characteristics tested
- The RFI detection algorithms tested
- The data sets generated for the test of the RFI detection algorithms
- Test Objectives
- Test Methods
- Assumptions
- Test Equipment Identification
- Environmental and/or other conditions to be maintained
- Test durations
- Test configurations
- Sequence of events
- Step-by-step instructions
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Analysis techniques
- Expected results
- Pass/Fail Criteria
- Test Termination Procedure

2. For each analysis required to validate the RFI detection capability tests, the procedure for each analysis shall include:

- Analysis Objectives, i.e., short description of the item or items being validated by this analysis
- Analysis Methods
- Assumptions
- Applicable Documents and Software
- Associated algorithms and software tools
- List of input data that will be used and related test IDs that were used to acquire the data
- List of external reference sources of input data and/or model used in analysis
- List of fixed analysis parameters and set points
- List of adjustable analysis parameters and set points that may vary between runs.
- Sequence of events
- Step-by-step description of the data analysis procedure/algorithm
- Data Recording/Output Format Requirements
- Data Recording Forms and Tables
- Expected results
- Pass/Fail Criteria

Raw data generated to assess the SMBA instrument's RFI detection capability for each test performed shall be delivered along with the test report and the analysis report. The raw data delivery shall include:

- The ancillary data used for the test
- Raw data
- Frequency channels, their bandwidth
- Implementation of the RFI detection algorithms tested

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-7

2. Title:INSTRUMENT CONCEPT OF OPERATIONS
DOCUMENT3. Reference:

SOW 2.2

4. Use:

To provide information on the Instrument Concept of Operations derived from unique aspects of the Contractor's design.

5. Preparation Information:

The Instrument Concept of Operations Document shall describe the methodology required to operate the instrument. Relative to the Contractor's design (including any and all instrument unique operational features), the operations concepts shall address:

- a) Instrument operational environment and limitations
- b) Operational requirements over and above existing instrument requirements necessary to accommodate design unique factors
- c) Unique ground command and data handling operations
- d) User support operations
- e) Contingency and emergency instrument unique operational scenarios, including flight software changes.
- f) RFI detection related activities

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-8

2. Title:Inputs to the Spacecraft to Instrument Interface
Control Document3. Reference:

SOW 2.2.2.1, 2.7, 6.6.2.3

4. Use:

To establish interface definition and control between SMBA and the spacecraft.

5. Preparation Information:

This document shall provide detailed technical information regarding the interface of the SMBA to the spacecraft. The inputs to the Spacecraft to Instrument ICD shall include interface requirements for the SMBA. Interface inputs shall consist of the following:

- ICD Agreements/Understanding
 - Flight or test hardware supplied by one organization to the other organization (e.g., matching connectors, test fixtures, GSE)
 - Services and materials provided by one organization to the other organization (e.g., purge gas and fittings)
 - Flight hardware design and fabrication responsibility for interfaces (e.g., flight harness segments, ground straps, fasteners)
 - Pointing budgets and allocations, including estimated required spacecraft allocations
- ICD Design
 - First circuit schematics for all power and signal interfaces
 - Connector I/O pinout tables (pin number and function) and harness wiring (e.g., twisted-shielded pair; 100 ohm controlled impedance)
 - CBE or measured orbit average power, by mode
 - CBE or measured Main Bus peak current
 - Modeled or measured Main Bus inrush transients
 - Modeled Main Bus impedance for source and load
 - Main Bus power service fault protection implementation (e.g., overcurrent set point)
 - Data – Rates (peak and orbit average), volume, compression, format, high speed data interface
 - Heater circuit schematic with equivalent resistance

-
- Design implementation of data bus and command/telemetry bus timing and protocols
 - Baseplate temperature range
 - Instrument calibration spacecraft requirements
 - The type, location, and orientation of the Instrument Alignment References
 - Mechanical
 - Mass, CG, MOI
 - All instrument fields of view, including, where applicable, optical, glint, thermal cryo-radiator, electronics radiator, and solar diffuser fields of view.
 - Static and dynamic envelopes/volumes and keep out zones, including thermal blankets, instrument connectors, mounting feet, instrument-supplied harnesses, and unique mounts.
 - Information on field-of-view-related issues such as clearances, multi-path tolerances, indirect effects such as glint and allowable secondary radiation, and percent interference allowed.
 - Engineering drawings with a set of "not to exceed" dimensions for the launch, deployed on-orbit, and installation envelopes.
 - Specific requirements for pinning the instrument.
 - The spacecraft mounting interface requirements for each separately-mountable instrument component.
 - Mounting hole pattern
 - Mounting footprint
 - Mounting fasteners
 - Mounting type and materials (related to thermal and electrical bonding)
 - Volume/envelop with dimensions
 - Ground strap attach points, if any
 - Connector location with connector designators
 - Fields of View
 - Harness and GSE restraints, support, mounting interfaces
 - Purge gas interface
 - Location and proximity of other hardware mounted on the same panel
 - Thermal
 - Interface temperatures, radiator FOV
 - Thermal Control System
 - Flight Software (FSW)
 - Protocol, rates, and handshaking (commanding acknowledgment)
 - Continuous, burst, etc.
 - FSW loads/patches for software updates
 - SMBA FSW updates
 - Operations
 - Operational modes: configuration, power draw, data rate, etc.
 - Calibration maneuvers
 - On-Orbit Constraints, Restrictions, and Warnings
 - Ground support Equipment used during observatory I&T
 - Mechanical Fixtures
 - Lifting Slings

- Simulators & Stimulators
- Contamination
 - Allocation of contaminant contributions from Observatory sources
 - Observatory I&T and launch site facility constraints
 - MLI vent locations
 - Budget for particulate and molecular from delivery through EOL
 - Purge requirements (if any)
- Observatory level instrument testing
 - Instrument simulators
 - I&T Constraints, Restrictions, and Warnings

The inputs to the ICD shall include the special sequences required to re-configure from sudden entry into OFF mode.

The inputs to the ICD shall include what activation sequences/events are required, where they are necessary, and specific spacecraft control requirements, including appropriate algorithms to be implemented.

The inputs shall include a list of items that may need to be accessible at observatory I&T, as well as a list of items to remove/install prior to flight, venting location, size, path, etc.

The Instrument contractor shall provide the location, size, path, venting forces and impulses, direction of net thrust, and operation time of vents in the instrument.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-9

2. Title:Inputs to the Mechanical Instrument Interface
Control Document3. Reference:

SOW 2.2.2.1

4. Use:

To establish interface the mechanical interface definition and control between SMBA and the spacecraft.

5. Preparation Information:

The Spacecraft to Instrument MICD shall contain, at a minimum, the following information:

- a. Instrument placement on Satellite
- b. Footprints
- c. Coordinate system
- d. Mass and center-of-gravity
- e. Volume allocation and clearances
- f. I&T access/clearances
- g. Alignment features
- h. Flight configuration (stowed for launch), to include with respect to fields of view and surrounding structure, components, and other hardware
- i. On-orbit configurations (each state and intermediate positions of moving mechanical assemblies and deployables), to include with respect to fields of view and surrounding structure, components, and other hardware
- j. Hardpoints and mounting interfaces for mechanical ground support equipment
- k. Fields-of-view (optical, thermal, glint, RF), including tolerances
- l. Electrical connector designators (e.g., J1, J2)
- m. Ground straps
- n. Mounting hardware (and provider), torque specifications

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-10

2. Title:Field Programmable Gate Array (FPGA) Design
Data Package.3. Reference:

SOW 4.2

4. Use:

An FPGA Design and Analysis Report is required for each FPGA design to allow Government review of Contractor FPGA designs.

5. Preparation Information:

An FPGA Design Data Package shall be delivered for all FPGA designs (heritage, modified, or newly developed). If an FPGA is revised after its Design Data Package was delivered, then the Contractor shall deliver an updated Design Data Package so that each FPGA's final Design Data Package contains the flight design.

NOTE: For heritage or inherited FPGA designs, it is expected that items (a) through (g) below will be delivered to the extent that these artifacts exist. It is not intended for the Contractor to create new artifacts for heritage or inherited designs simply for submission with this CDRL/DID.

The FPGA Design Data Package shall include:

- a. Design (place and route) database
- b. Synthesis report files
- c. Circuit board schematics containing this gate array
- d. Circuit board netlist(s) (any ASCII format such as PADS, MGC, Allegro)
- e. Requirements, specifications, and verification document(s) as well as any supporting material (e.g., block diagrams, presentation material) relevant to the gate array
- f. Source code (e.g., VHDL, Verilog, state machines, state tables Portable Document Format (PDF) of schematics) including any third party intellectual property code and cores
- g. Responses to the GSFC 500-PG-8700-2.7B, Design of Space Flight Field Programmable Gate Arrays checklist or equivalent including:
 - a. Timing analyses for external inputs and outputs, internal domain(s), etc.
 - b. Disposition of all clock domain crossings.

- c. Simulation analysis summary (e.g., functional coverage, code coverage) and simulation test bench/script code.
- d. Signal integrity analyses relevant to this gate array
- e. Power integrity analyses relevant to this gate array

To facilitate reviewing the FPGA design, the FPGA Design Data Package may also include:

- a. System, box, and circuit board requirements, specifications, presentations, and verification document(s) relevant to the gate array and its role in the system, box, and circuit board
- b. Board part list (any ASCII or common spreadsheet format)
- c. PDF of the board layout, such as an assembly drawing
- d. Any other files relevant to the review of the gate array

For any heritage, inherited, or build-to-print FPGA designs, the FPGA Design Data Package shall also include a “was/is” table consisting of:

- a. Full manufacturer part number
- b. Lot/Date code(s)
- c. Vendor part number (e.g., Altered Item Drawing number, etc.)
- d. Logic (e.g., any changes in the hardware description language code, etc.)
- e. Synthesis (e.g., not re-synthesized, re-synthesized to preclude Government Industry Data Exchange Program (GIDEP) xyz, etc.)
- f. Layout (e.g., not re-routed, re-routed per GIDEP xyz, etc.)
- g. Verifications
 - a. Simulations (e.g., not re-run, new simulations added, etc.)
 - b. Analyses (e.g., not updated, timing analysis updated with values exported from latest software version that supports this FPGA, etc.)
 - c. Tests
- h. Environments
 - a. Operating temperature range
 - b. Operating voltage range(s)
 - c. Radiation
- i. Changes in manufacturing process and/or foundry

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-11

2. Title:

MASS PROPERTIES REPORT

3. Reference:

SOW 2.2

4. Use:

The purpose of the Mass Properties Report is to document all mass properties of the Instrument, its subsystems, and components from preliminary design through final assembly and observatory integration, and to satisfy the mass properties reporting requirements of the launch services provider.

5. Preparation Information:

This document shall provide a mass properties database for Instrument. Mass properties shall include mass, center of gravity, moments of inertia, products of inertia, uncompensated momentum, principal axis misalignment, relevant alignment measurements for each separately-mounted component of the instrument, for each delivered unit, as well a composite total, and physical dimensions, where applicable. The report shall be initially based upon calculated values and shall be updated as calculations are revised and actual measured data becomes available. Following environmental testing and prior to the Instrument shipment to the observatory integrator, the report shall contain a complete mass properties summary of the final Instrument mass properties as measured. The report shall also include the appropriate mass contingency for the current stage of hardware development, along with the allocated mass allowables. If the instrument contains movable masses, expendable masses, or deployables, the respective variations shall be documented in the report.

The mass properties report shall contain the following:

- a. An overall Instrument mass summary. In addition provide: (a) An Instrument mass properties summary for the various phases of the mission through Observatory integration. (b) A detailed mass properties summary of all Instrument hardware organized by subsystem. (c) A summary of all mass properties changes incorporated into the Instrument mass properties database since the last report.

The deliveries at L-54 weeks and L-20 weeks shall satisfy launch services provider requirements as they apply to Instrument, and shall include nominal and 3-sigma uncertainties for mass, centers of gravity, moments of inertia, products of inertia, and principal axis misalignment.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-12

2. Title:

SPARE PARTS PLAN AND LIST

3. Reference:

SOW 4.1, 4.3.2.3

4. Use:

To review Contractor's spares plan and parts list.

5. Preparation Information:

The Spares Parts Plan shall define and justify the Contractor's position for the spares proposed for the Instrument build, preparation, and test. This plan shall also present the schedule and method for obtaining the spares. The Plan shall provide a listing of Spare Parts. For the purposes of this CDRL, the Contractor should concentrate on parts that are not commonly available, or may cause schedule problems if out of stock.

The Contractor shall develop, deliver, and implement an approved spare parts program plan for critical flight spares and for critical Ground Support Equipment (GSE) spares built by the Contractor (e.g., Calibration Test Equipment, target assemblies and electronics). The plan shall consider the judicious definition, procurement, and stock of a set of these critical spares to help minimize potential schedule impact created by subsystem failures, by contamination, or by other plausible events or conditions.

Once the plan is approved by the Government, the Contractor shall procure sufficient spare parts to account for typical attrition during the manufacturing process due to handling, random failure, programming, etc. The Contractor shall provide a spare kit of electrical, electronic, electro-mechanical (EEE) parts and spare boards. The spare boards shall have coupons evaluated by the Government. The spares may be retained as fully assembled, partially assembled or piece part spares. Since the need to use a spare may be a failure condition in the instrument and, depending on the specifics of the fault, the spare itself may need rework or repair, simple substitution of a spare may not be a viable approach. With this consideration in mind, the judgment of the Contractor shall establish the assembly level of sparing.

In defining the spare parts program, the Contractor shall consider the reliability, handling, and environment conditions of subsystems, components, and parts under consideration for sparing, including the likelihood that these items would need to be replaced. The Contractor should provide enough spares that they could recover from a failure within three months. The spare

parts shall be qualified, tested, and calibrated to the same standards as the flight instrument articles or primary GSE article consistent with their spared level of assembly.

DATA ITEM DESCRIPTION

1. CDRL No.:

SE-13

2. Title:

FIRMWARE DESIGN DATA PACKAGE

3. Reference:

SOW 2.0

4. Use:

A Firmware Design Data Package is required for each firmware design to allow independent Government review and assessment of Contractor firmware designs.

5. Preparation Information:

The final, flight Firmware Design Data Package shall be submitted for each Firmware developed by the Contractor. If the flight version is not ready for review when requested by the Government, a Design Data Package for the current engineering/test/prototype Firmware shall be submitted as well.

A Firmware Design Data Package shall include the:

1. Firmware design (place and route) database and any constraint file(s)
 - a. For Actel SX and AX FPGAs, the database file is the .adb file
2. Firmware synthesis constraint and report files
 - a. For designs synthesized with Synplify, the synthesis report is the .srr file
3. Firmware timing analyses for external inputs and outputs, internal domain(s), etc.
4. Firmware internal clock domain crossings disposition table/spreadsheet
 - a. For each clock domain crossing, briefly describe how the source and destination domain logic work without error
5. Firmware simulation coverage report and testbenches/scripts
6. Firmware source code (e.g., Very High-Speed Integrated Circuits Hardware Description Language (VHDL), Verilog), schematics and/or state machines/tables, including any third party intellectual property code and/or cores
7. System, box, board & Firmware requirements, specifications presentations and/or verification document(s) relevant to the Firmware and its role in the system, box, board
8. Board schematics (PDF and native format) that contain this Firmware
9. Board netlist (any ASCII format)
10. Board part list (any ASCII or common spreadsheet format)
11. Board layout (PDF and native format)
12. Signal integrity analyses relevant to this Firmware

Appendix A: Abbreviations and Acronyms

ΔG/G	Gain Stability
A	Approval
ACS	Attitude Control System
ADB	Ada body file extension
AFM	Additional Flight Models (i.e., in addition to FM1)
AM	Additive Manufacturing
AMCP	Additive Manufacturing Control Plan
ANSI	American National Standards Institute
APID	Application Process Identifier
APPLIC	Applicability
ASCII	American Code Standard for Information Interchange
ATMS	Advanced Technology Microwave Sounder
ATP	Ambient Temperature and Pressure
ATP	Authority To Proceed
AVI	Audio Video Interleave
BAT	Bench Acceptance Test
BCR	Baseline Change Request
BOL	Beginning of Life
C&DH	Command and Data Handling
C&T	Command and Telemetry
CADRe	Cost Analysis Data Requirement
CAD	Computer-Aided Design
CAE	Computer-Aided Engineering
Cal/Val	Calibration/Validation
CAT	Category
CBE	Current Best Estimate
CCB	Configuration Control Board
CCP	Contamination Control Plan
CCR	Configuration Change Request
CDR	Critical Design Review
CDRL	Contract Data Requirements List
CG	Center of Gravity
CIIL	Configuration Item Identification List
CIL	Critical Items List
CM	Configuration Management
CMO	Configuration Management Office
CMP	Configuration Management Plan
COR	Contracting Officer's Representative
COTS	Commercial-off-the-shelf
CPR	Contract Performance Report
CPT	Comprehensive Performance Test
CPU	Central Processing Unit
CRC	Cyclic Redundancy Check

CrIS	Cross-track Infrared Sounder
CSC	Computer Software Components
CSCI	Computer Software Configuration Item
CSU	Computer Software Unit
csv	Comma Separated Values file extension
CTB	Consent to Break
CTDB	Command and Telemetry Database
CTI	Consent to Integrate
CTP	Consent to Proceed
CV	Calibration/Validation
D	Definitization
DACA	Days After Contract Award
DC	Direct Current
dCDR	Delta Critical Design Review
DD	Department of Defense (forms only)
Del	Deliverable
DFMR	Designed For Minimum Risk
DID	Data Item Description
DM	Data manager
DMR	Design Modification Review
DPA	Destructive Physical Analysis
DOORS	Dynamic Object-Oriented Requirements System
DTD	Document Type Definition
DTM	Detailed Thermal Model
DWF	Design Web Format
E	Electronic Format
EAC	Estimate at Completion
EAR	Export Administration Regulations
EDR	Environmental Data Records
EDU	Engineering Development Unit
EEE	Electrical, Electronic, and Electromechanical
EEO	Early Engineering Opportunity
EGSE	Electrical Ground Support Equipment
EIDP	End Item Data Package
EM	Electromagnetic
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EOL	End of Life
EOMP	End of Mission Plan
EPR	Engineering Peer Review
ESD	Electrostatic Discharge
ESRD	Electrical System Requirements Document
ETU	Engineering Test Unit
EVMIG	Earned Value Management Implementation Guide
EVM(S)	Earned Value Management (System)

EVP	Environmental Verification Plan
FAR	Federal Acquisition Regulation
FCA	Functional Configuration Audit
FEM	Finite Element Model
FM	Flight Model
FMEA	Failure Modes and Effects Analysis
FMECA	Failure Modes and Effects Criticality Analysis
FOV	Field of View
FPGA	Field Programmable Gate Array
FQT	Formal Qualification Test
FRB	Failure Review Board
FRR	Flight Readiness Review
FSTB	Flight Software Test Bed
FSW	Flight Software
FVTS	Flight Vehicle Test Suite
G-R	Giver-Receiver
GFE	Government Furnished Equipment
GIDEP	Government Industry Data Exchange Program
GOLD	GSFC Open Learning Design
GOTS	Government-off-the-shelf
GPR	Goddard Procedural Requirement
GSE	Ground Support Equipment
GSFC	Goddard Space Flight Center
H	Hardcopy
H&S	Hardness and Survivability
Hz	Hertz
I	Information
I/O	Input/Output
I&T	Integration and Test
ICDR	Instrument CDR
ID	Identification
IBR	Integrated Baseline Review
ICD	Interface Control Document
ICDR	Instrument Critical Design Review
iDel	Internal Deliverable
IEEE	Institute of Electrical and Electronics Engineers
IF	Intermediate Frequency
IMAR	Instrument Mission Assurance Requirements
IMS	Integrated Master Schedule
IMT	Integrated Mission Timeline
IOC	Initial Operating Capability
IPMR	Integrated Program Management Report
IPT	Integrated Product Team
IR	Infrared
IRD	Interface Requirement Document

iRec	Internal Receivable
ISAR	Instrument Safety Assessment Report
ISRR	Instrument System Readiness Review
IT	Integration and Test
ITAR	International Traffic in Arms Regulations
IV&V	Independent Verification and Validation
JCT	Joint Compatibility Test
JPEG	Joint Photographic Experts Group
JPSS	Joint Polar Satellite System
K	Kelvin
KDP	Key Decision Point
Kg	kilograms
L	Launch
LEO	Low Earth Orbit
LEO&A	Launch, Early Orbit, and Activation
LOA	Letter of Agreement
LPD	Low Earth Orbit Programs Division
LPT	Limited Performance Test
LRD	Launch Readiness Date
LRR	Launch Readiness Review
M&P	Materials and Processes
MA	Mission Assurance
MATLAB	Matrix Laboratory
Mbps	Mega bits per second
MBSE	Model Based Systems Engineering
MDR	Mission Definition Review
MGC	ASCII file format extension
MGSE	Mechanical Ground Support Equipment
MICD	Mechanical Interface Control Document
MIUL	Materials Identification and Usage List
MIP	Mandatory Inspection Point
MLI	Multi-Layer Insulation
MOC	Mission Operations Center
MOI	Moment of Inertia
MOR	Mission Operations Review
MOT	Mission Operations Team
MOTS	Modified-off-the-shelf
MOU	Memorandum of Understanding
MPEG	Moving Picture Experts Group
MPSCIP	Materials and Processes Selection, Control, and Implementation Plan
MR	Mission Rehearsal
MRB	Material Review Board
MRR	Manufacturing Readiness Review or Mission Readiness Review
MST	Mission Support Team
MUA	Materials Usage Agreement

NASA	National Aeronautics and Space Administration
NASTRAN	NASA Structural Analysis
NEAT or NEdT	Noise Equivalent Delta Temperature
NEON	Near Earth Orbit Network
NF	Noise Factor
NFS	NASA FAR Supplement
NGS	NOAA Ground System
NIST	National Institute of Standards and Technology
NLT	No Later Than
NOAA	National Oceanic and Atmospheric Administration
NPR	NASA Procedural Requirement
NS1	NEON Series-1 Project
NSOF	NOAA Satellite Operations Facility
OCDR	Observatory Critical Design Review
ODAR	Orbital Debris Assessment Report
OHA	Operations Hazard Analysis
OHR	Operations Handover Review
OO	On Orbit
P _{CS}	Probability of not meeting cost/schedule allocations
P _S	Probability of safety event occurrences
P _T	Probability of not meeting technical performance requirements
PADS	ASCII file format from Mentor Graphics
PCA	Physical Configuration Audit
PCB	Printed Circuit Board
PDF	Portable Data Format
PDR	Preliminary Design Review
PDS	Post-Delivery Support
PER	Pre-Environmental Review
PHS&T	Packaging, Handling, Storage, and Transportation
PLAR	Post-Launch Assessment Review
PLO	Phase-Locked Oscillator
PMB	Performance Measurement Baseline
PM&P	Parts, Materials and Processes
PMP	Project Management Plan
PMPCA	Parts, Materials and Process Control Authority
PMR	Program Management Review
PPP	Part Production Plan
PRD	Performance Requirements Document
PROC	Procedures
PROM	Programmable Only Memory
PRT	Platinum Resistance Thermometer
PSR	Instrument Pre-Ship Review
PSR	Program Status Review
QA	Quality Assurance

QPMR	Quarterly Program Management Review
R	For Review
RAM	Random Access Memory
RDR	Raw Data Record
RE	Review
Rec	Receivable
Rec-Del	Receive-Deliver
RF	Radio Frequency
RFE	Receiver Front End
RFA	Request for Action
RFI	Radio Frequency Interference
RHA	Radiation Hardness Assurance
RMM	Radiometric Math Model
RMP	Risk Management Plan
ROP	Recommended Operating Procedures
RPS	Receiver Power Supply
RTM	Reduced Thermal Model
RVM	Requirement Verification Matrix
SC	Spacecraft
SCCDR	Spacecraft CDR
SCPDR	Spacecraft PDR
SCIRD	Spacecraft IRD
SDD	Software Detailed Design
SDE	Scan Drive Electronics
SDP	Software Development Plan
SDR	Sensor Data Record
SDR	System Definition Review
SE	Systems Engine
SEE	Single Event Effect
SEMP	Systems Engineering Management Plan
SEN	Software Engineering Notebook
SET	Single Event Transient
SFTP	Secure File Transfer Protocol
SIR	System Integration Review
SMBA	Sounder for Microwave-Based Applications
SOE	Satellite Operations Exercise
SOW	Statement of Work
SPA	Signal Processor Assembly
SPMM	Sensor Performance Math Model
SPVM	System Performance Verification Matrix
SPVP	System Performance Verification Plan
SRF	Spectral Response Function
SRR	Systems Requirements Review
SRR	Scene Release Reconstruction data file extension
STE	System Test Equipment

STEP	Standard for the Exchange of Product Data
SW	Software
SWAR	Software Acceptance Review
SWE	Software Engineering Requirement
SWTRR	Software Test Readiness Review
T	Time of the launch date
TAA	Technical Assistance Agreement
TBD	To Be Determined
TBR	To Be Resolved
TBS	To Be Scheduled
TDA	Test Data Analysis
TDR	Temperature Data Record
TID	Total Ionizing Dose
TM	Torque Margin
TMM	Thermal Math Model
TPM	Technical Performance Metric
TQCM	Thermal Quartz Crystal Microbalance
TRL	Technology Readiness Level
TRR	Test Readiness Review
TVAC	Thermal Vacuum
TYF	Test as You Fly
UID	User Identification
U.S.	United States
VDD	Version Description Document
VHDL	Very High-Speed Integrated Circuits Hardware Description Language
VTL	Verification Tracking Log
W	Watt
WBS	Work Breakdown Structure
WGVC	Working Group on Calibration and Validation
XML	Extensible Markup Language